

A virally encoded tRNA neutralizes the PARIS antiviral defence system

<https://doi.org/10.1038/s41586-024-07874-3>

Received: 22 December 2023

Accepted: 24 July 2024

Published online: 7 August 2024

Open access

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Viruses compete with each other for limited cellular resources, and some deliver defence mechanisms that protect the host from competing genetic parasites¹. The phage antirestriction induced system (PARIS) is a defence system, often encoded in viral genomes, that is composed of a 55 kDa ABC ATPase (AriA) and a 35 kDa TOPRIM nuclease (AriB)². However, the mechanism by which AriA and AriB function in phage defence is unknown. Here we show that AriA and AriB assemble into a 425 kDa supramolecular immune complex. We use cryo-electron microscopy to determine the structure of this complex, thereby explaining how six molecules of AriA assemble into a propeller-shaped scaffold that coordinates three subunits of AriB. ATP-dependent detection of foreign proteins triggers the release of AriB, which assembles into a homodimeric nuclease that blocks infection by cleaving host lysine transfer RNA. Phage T5 subverts PARIS immunity through expression of a lysine transfer RNA variant that is not cleaved by PARIS, thereby restoring viral infection. Collectively, these data explain how AriA functions as an ATP-dependent sensor that detects viral proteins and activates the AriB toxin. PARIS is one of an emerging set of immune systems that form macromolecular complexes for the recognition of foreign proteins, rather than foreign nucleic acids³.

Antiviral defence systems in bacteria and archaea are extraordinarily diverse and many are mechanistically similar to immune responses in eukaryotic cells⁴. The recent expansion of bacterial and archaeal immune systems stems from the appreciation that they tend to colocalize in the genome⁵ and are often carried by mobile genetic elements, including prophages and satellite viruses⁶. The phage antirestriction induced system (PARIS) is one of several defence systems recently identified within a hotspot of genetic diversity carried by P4 phage satellites². The PARIS system of *Escherichia coli* strain B185 protects against phage T7 infection through a mechanism triggered by sensing the T7 overcoming classical restriction (Ocr) protein. Ocr is a DNA mimic that inhibits type I restriction-modification and BREX defence systems. Thus PARIS is an ‘anti-antirestriction’ system that senses a viral immune suppressor.

The PARIS defence system consists of two proteins, AriA and AriB (Fig. 1a,b). AriA contains a conserved ABC ATPase domain whereas

AriB is a domain of unknown function (DUF4435), which includes a TOPRIM nuclease^{7,8}. AriA and AriB are usually separate proteins, although they are occasionally fused into a single polypeptide. Several defence systems share ABC ATPase and TOPRIM domains, including the protein-overcoming lysogenization defect (OLD) produced by the P2 prophage, which blocks infection by phage lambda⁹. OLD does not directly sense a phage protein, but rather inactivation of the RecBCD exonuclease¹⁰. Activation of OLD results in translation inhibition⁹, but the underlying mechanism has not been determined.

Association between the ABC ATPase and TOPRIM domains is an emerging theme in diverse recently discovered immune systems¹¹. Examples include the Gabija system^{12,13}, effector toxin proteins of ret-ron immunity¹⁴ and bona fide toxin–antitoxin modules¹⁵. This implies a broad adaptability of the conserved ABC ATPase and TOPRIM nuclease architecture to a variety of antigenic stimuli, resulting in diversification of the immune modules probably acting through abortive mechanisms.

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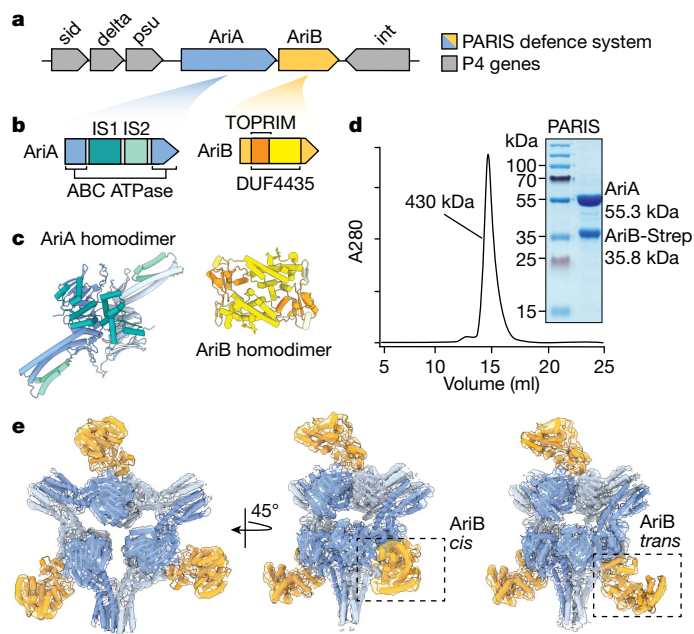


Fig. 1 | PARIS is a two-component system that assembles into a propeller-shaped supramolecular complex. **a**, Schematic of the integrated P4 satellite in the *E. coli* genome. Grey arrows labelled *sid* (size determination), *delta* (δ transcriptional activator), *psu* (polarity suppression) and *int* (integrase) are P4 genes flanking *AriA* and *AriB* genes. **b**, PARIS genes coloured according to domain organization. *AriA* is an ABC ATPase whereas *AriB* is a DUF4435 that includes a TOPPRM nuclease. IS1 is composed of three helices flanked by the ABC ATPase domain, whereas IS2 adds two additional helices to the coiled-coil resulting in a helical bundle. **c**, AlphaFold2-predicted structures of *AriA* and *AriB* as homodimers coloured according to the structural features shown in **b**. **d**, Size exclusion chromatography showing that *AriA* and *AriB* copurify as a complex of roughly 430 kDa. The size exclusion chromatography profile is representative of three independent biological experiments. **e**, Experimentally determined density maps of the PARIS complex showing a homohexameric assembly of *AriA* with D3 symmetry. The *AriA* hexamer is decorated by *AriB* subunits that attach to *AriA* homodimers in one of two potential orientations: a 'cis' arrangement where the three *AriB* subunits are facing the same direction, or a 'trans' arrangement where one *AriB* subunit is rotated by 180° relative to the other two.

We currently have little understanding of the molecular mechanisms underpinning how these systems sense and respond to viral infections.

Here we use cryo-electron microscopy (cryo-EM) to show that the *E. coli* B185 PARIS system components assemble into a 425 kDa, propeller-shaped complex with six molecules of *AriA* and three subunits of *AriB*. We show that *AriA* senses the T7 Ocr antirestriction protein leading to the release of the *AriB* nuclease effector. Release of *AriB* from the *AriA* scaffold is necessary for potentiation of *AriB*, which forms a nuclease-active dimer that degrades host lysine transfer RNA (tRNA^{Lys}), thereby blocking translation and leading to growth arrest or cell death. We find that phage T5 carries its own tRNA^{Lys} isoacceptor that rescues the phage from PARIS-mediated defence. Finally, we perform a phylogenetic analysis of PARIS systems that provides insights into their evolutionary history and shows how they relate to other defence systems that use ABC ATPases.

PARIS forms a propeller-shaped complex

The PARIS defence system consists of two proteins, *AriA* and *AriB*, which protect cells from phages through a previously undetermined mechanism². Here, we focused on the system from the P4 satellite integrated into the *E. coli* (strain B185) genome (Fig. 1a,b). Based on published experimental structures of related proteins^{8,13} and AlphaFold2 (ref. 16)

predictions (Fig. 1c and Extended Data Fig. 1a–d), we anticipated that *AriA*/*AriB* would assemble into a multisubunit complex. To determine how *AriA* and *AriB* assemble, we overexpressed *AriA* with a C-terminally tagged *AriB* (*AriB*-Strep), pulled down on *AriB*, and purified the complex using size exclusion chromatography (Fig. 1d). The *AriA* and *AriB* proteins coelute in a single peak with an estimated molecular weight of 430 kDa. Next, we used cryo-EM to determine the structures of the PARIS complex (Fig. 1e). The structure explains how six molecules of *AriA* combine to form a D3 symmetric scaffold decorated by three subunits of *AriB*. The helical domains, which typically provide a platform for DNA binding in related ABC ATPases¹⁷, instead function as additional dimerization interfaces that enable a trimeric assembly of *AriA* homodimers (Fig. 1c and Extended Data Fig. 2). Dimerization of the helical domains from adjacent *AriA* homodimers form three blades of the propeller (Fig. 1e and Extended Data Fig. 2). Additional density is evident between each of the *AriA* blades, where *AriB* attaches to *AriA* near the ATP-binding sites. However, density in this region of the reconstruction shows conformational heterogeneity, consistent with *AriB* attaching to either, but not both, *AriA* molecules at the dimer interface (Extended Data Fig. 1e,f and Supplementary Video 1). Using multiple rounds of focused three-dimensional classification, we resolved two distinct isomers of the complex (Fig. 1e and Extended Data Fig. 3d). In one reconstruction, *AriB* attaches to three symmetrically equivalent *AriA* molecules (3.7 Å resolution, *cis* configuration) and, in the other, one of the three *AriB* molecules binds the opposing *AriA*, which flips *AriB* 180° relative to the other two *AriB* subunits (3.9 Å resolution, *trans* configuration; Extended Data Fig. 3g,h and Supplementary Video 1).

AriA sequesters *AriB*

Although the homohexameric assembly of *AriA* was apparent in the early stages of reconstruction, conformational heterogeneity limited resolution (Extended Data Fig. 3 and Supplementary Video 1). To improve resolution, we used local refinement to determine a 3.2 Å nominal resolution reconstruction of one asymmetric unit of the PARIS complex (*AriA*₂-*AriB*₁), into which we built an atomic model (PDB: 8UX9; Fig. 2 and Extended Data Table 1). Each asymmetric unit of the PARIS complex is composed of two *AriA* molecules that form a head-to-tail homodimer^{18–20}, and one *AriB* molecule that binds directly above the *AriA* dimer interface.

AriB is a metal-dependent TOPPRM nuclease and the predicted effector of the PARIS defence system^{2,7,8}. The structure shows a series of spatially conserved, negatively charged residues (E26, D30, D88, E90 and E122) that are characteristic of TOPPRM domains that bind two metal ions⁷. The E26A mutation inactivates PARIS-mediated defence² (Fig. 2b). Glutamic acid 26 is just upstream of an N-terminal alpha helix (α 2) on *AriB*, which includes two arginine residues (R31 and R28) that form salt bridges with negatively charged residues on *AriA*, which is positioned directly above the ATP-binding sites (Fig. 2b, Extended Data Fig. 2 and Supplementary Video 1). Mutations that interfere with the *AriA*-*AriB* interface (*AriB* R28E, *AriB* R31E and *AriB* R31E/R28E or *AriA* D401N/E404Q) limit phage protection by PARIS (Fig. 2e), which suggests that the association between *AriA* and *AriB* is required to provide protection.

The *AriA* ABC ATPase domain resembles that of Rad50 (root mean squared deviation 1.1 Å across 86 C-alpha atom pairs), a universally conserved protein involved in double-stranded DNA break repair^{21–23}. Structural comparison of *AriA* with Rad50 confirms that *AriA* contains the sequence motifs required for ATPase activity, along with two unique insertion sequences, insertion sequences 1 and 2 (IS1 and IS2, respectively; Fig. 1b and Extended Data Fig. 2). Insertion sequence 2 contains a series of aromatic residues located on one face of helix α 8 (F230, F234, F238 and Y304) that coordinate interactions between *AriA* homodimers, and explains how *AriA* homodimers assemble into a trimer of dimers (Extended Data Fig. 2), a rare assembly state for this family of proteins²¹. Like other ABC ATPases, the head-to-tail assembly

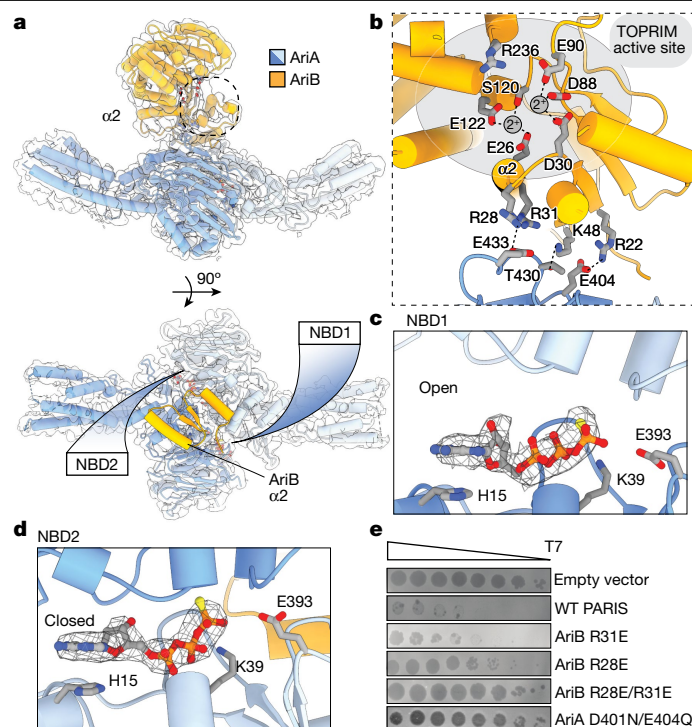


Fig. 2 | Interaction between AriA ATPase and AriB is essential for PARIS-mediated defence. **a**, Each asymmetric unit of the PARIS complex contains an AriA homodimer with two NBDs. A 90° rotation of the asymmetric unit, showing 1–53 N-terminal residues (yellow) of AriB positioned over the ATPase active sites of AriA. **b**, AriB attaches to AriA through electrostatic interactions near helix α2 of AriB. TOPRIM active site residues predicted to

participate in metal binding (grey spheres) and catalysis are shown as sticks. **c**, Close-up of NBD1 showing ATPγS bound in an open conformation. **d**, Close-up of NBD2 showing the bound ligand trapped in a closed precleavage state. **e**, Plaque assays were performed in duplicate with two independent clones of each mutant, using a serial dilution of phage T7 on *E. coli* MG1655 carrying the WT or mutated PARIS system.

of the AriA homodimer results in two symmetrically positioned nucleotide-binding domains (NBDs) at the dimer interface (Fig. 2a). In the presence of a non-hydrolysable ATP analogue (that is, ATPγS), we observe density for the ligand in both NBDs but in different coordination states (Fig. 2c,d). Previous studies of ABC ATPases showed that ATP hydrolysis occurs at the two NBDs in an alternating fashion^{21,24}, which is consistent with the observed ‘open’ and ‘closed’ conformations of NBDs (Fig. 2c,d and Supplementary Video 1).

AriA binds to Ocr and releases AriB

The T7 phage antirestriction protein Ocr was previously identified as a trigger of the PARIS system, and its expression in the presence of PARIS leads to growth arrest or cell death². To determine whether AriA–AriB senses Ocr through direct interaction, we coexpressed AriA with a non-toxic active site mutant of AriB (E26A) and Ocr-Strep. AriA, but not AriB, copurified with Ocr-Strep and the complex migrated according to the predicted mass of the AriA hexamer (Fig. 3a and Extended Data Fig. 5a,b). This is consistent with a direct interaction between AriA and Ocr and release of AriB. A pull-down with AriB-Strep confirmed that AriB dissociates from the complex in the presence of Ocr (Fig. 3b).

To determine how Ocr triggers the release of AriB, we used the purified AriA–Ocr-Strep complex for structure determination. The 4.4-Å-resolution structure shows a D3 symmetric, homohexamer of AriA with no AriB (Fig. 3c and Extended Data Fig. 6). At this resolution, the structure of AriA purified using Ocr-Strep is indistinguishable from that of AriA purified using AriB-Strep. We anticipated that the structure would include Ocr, because Ocr-Strep was used to pull down the AriA hexamer and the complex remained stable during gel filtration. However, multiple data-processing strategies failed to resolve Ocr, suggesting that Ocr has several binding sites on AriA and/or that the interaction is conformationally flexible.

Because Ocr is a negatively charged DNA mimic^{25,26}, we hypothesized that the binding site on AriA would be positively charged. An electrostatic surface of the AriA hexamer shows a central pore, flanked by a series of three positively charged radial pores (Fig. 3c). These radial pores contain 2 flexible loops (one from each of the adjacent AriA homodimers) with 14 unmodelled residues (161–174, SSDVGYERR-VIRSS), with the central pore containing 6 loops (one from each AriA) incorporating 12 unmodelled residues (113–124, ESERHLRERDVK). To determine the role of these loops in phage defence, we tested the effects of alanine substitutions and charge swap mutations on positively charged residues. Arginine-to-alanine substitutions (R116A and R119A) in the central pore resulted in partial loss of phage protection (Fig. 3d). Attempts to generate other mutations in the central (R116E, R119E) or radial pores (R168E, R168A, R172E or R172A) of AriA consistently failed when we used a vector that also included wild-type (WT) AriB. We hypothesized that these mutations in AriA mimic trigger binding, which releases AriB and results in toxicity. To test this hypothesis, we expressed the central pore mutant (R116E/R119E) with the catalytically inactive AriB (E26A), which is non-toxic. Pull-down experiments performed with Ocr-Strep (Fig. 3e) or AriB-Strep (Fig. 3f) demonstrated that AriA (R116E/R119E) has limited interactions with both Ocr and AriB. Collectively, these results suggest that either the central or radial pores, or both, are involved in interaction with the trigger.

AriB alone does not provide phage defence and is only mildly toxic when overexpressed² (Extended Data Fig. 4). However, activated AriB (that is, AriB released from PARIS) is significantly more toxic. Moreover, mutations in AriA that disrupt interaction with AriB limit AriB toxicity and abolish phage defence. Collectively, these data suggest that the toxicity of AriB depends on its initial association with AriA.

The ATPase activity of AriA is also required for phage defence², which suggests that the ATPase of AriA is necessary for loading of AriB onto AriA, or release of activated AriB. To differentiate between loading or release

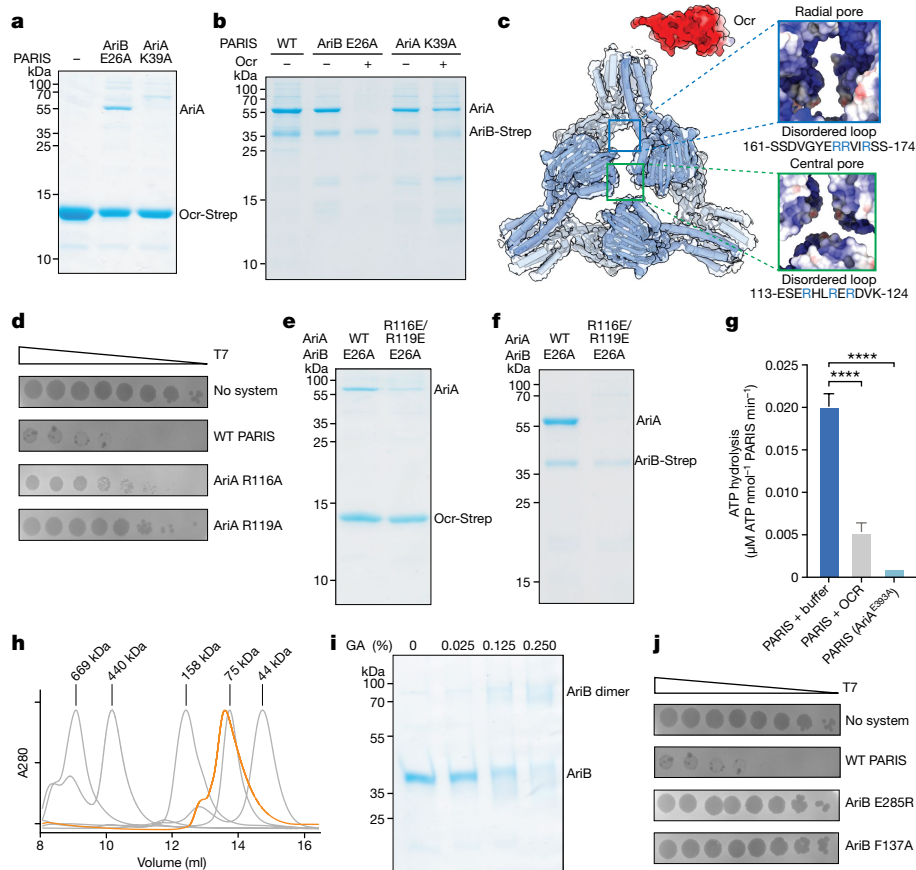


Fig. 3 | AriA interacts with Ocr in an ATPase-dependent manner, leading to the release of AriB from the PARIS complex. **a**, Ocr-Strep pulls down AriA, but not AriB. A mutation in the ATPase active site of AriA (K39A) blocks binding to Ocr-Strep. **b**, Ocr releases AriB (E26A) from WT AriA (lane 3), but Ocr does not trigger AriB release from an ATP-binding-deficient variant of AriA (K39A) (lane 5). **c**, A 4.4-Å-resolution reconstruction of AriA that was purified using Ocr-Strep as a bait. The structure shows three radial pores symmetrically positioned around a central pore. The pores contain disordered loops with several positively charged residues. Ocr (PDB: 1S7Z) is shown as an electrostatic surface. **d**, Mutations in the central pore (R116A or R119A) reduce the efficiency of phage protection. **e**, Ocr-Strep binds WT AriA and ejects AriB (E26A), and the charge swap mutations in the central pore of AriA (R116E/R119E) limit interaction with Ocr-Strep. **f**, AriB-Strep (E26A) pulls down WT AriA but not

AriA (R116E/R119E). **g**, ATP and ADP were separated using thin-layer chromatography (Extended Data Fig. 7), and the rates of ATP hydrolysis for PARIS, PARIS mixed with tenfold excess Ocr and PARIS with an ATPase mutation in AriA (E393A) were measured. Experiments were performed in triplicate, and error bars show $\pm$ s.e.m. Two-sided statistical analysis performed using a post hoc Dunnett's test, **** $P < 0.0001$. **h**, Size exclusion chromatography of AriB-Strep (around 36 kDa) following Ocr-mediated release from AriA. The column was calibrated using molecular weight standards (grey lines). AriB elutes in a single peak with an estimated molecular weight of 81 kDa (Extended Data Fig. 5a). **i**, Glutaraldehyde (GA) cross-linking assay with activated AriB (E26A). **j**, Mutations predicted to block AriB dimerization (E285R and F137A) prevent PARIS-mediated defence (Extended Data Fig. 1g). Plaque assays were performed in duplicate with two independent clones of each mutant.

of AriB, we performed pull-down assays using AriB-Strep and WT or an ATP-binding-defective mutant of AriA (K39A in Walker A). AriB-Strep associates with mutant and WT AriA at similar efficiency (Fig. 3b), demonstrating that the ATPase mutant of AriA loads, but does not release, AriB in the presence of Ocr. We reasoned that the lack of AriB release in the presence of AriA K39A is due to a defect in trigger recognition. To confirm the role of ATP in AriA-mediated trigger recognition, we conducted pull-downs using Ocr-Strep and WT or K39A AriA. The AriA K39A mutant failed to copurify with Ocr, indicating that trigger recognition is ATP dependent (Fig. 3a). To determine whether or how the trigger impacts the ATPase activity of AriA, we measured the rate of PARIS-mediated ATP hydrolysis in the presence or absence of Ocr and compared these results with an ATPase-defective AriA mutant (E393A in Walker B; Fig. 3g). The results demonstrated that Ocr significantly ($P = 0.0001$) reduced ATP turnover by AriA (Fig. 3g and Extended Data Fig. 7).

TOPRIM-containing proteins are known to form homodimers^{27,28}, but a predicted homodimer of AriB generated using AlphaFold2 clashed with AriA (Extended Data Fig. 1f). Thus, we hypothesized that AriB forms an active homodimer following trigger-mediated release

from AriA. The toxicity of PARIS following activation makes purification of active AriB challenging. To overcome this problem, we mixed lysates from cells expressing AriA–AriB-Strep with lysates from cells expressing Ocr, and then recovered activated AriB using affinity chromatography (Extended Data Fig. 5c). Size exclusion chromatography indicated that activated AriB forms a homodimer (Fig. 3h and Extended Data Fig. 5a), which was further confirmed by glutaraldehyde cross-linking experiments (Fig. 3i). Mutations in the AriB dimerization interface (E285R and F137A), as predicted from the AlphaFold2 model (Fig. 1c and Extended Data Fig. 1g), resulted in the loss of phage defence (Fig. 3j), highlighting that AriB dimerization is essential for PARIS activity. Collectively, these data show that the Ocr trigger interacts with AriA in an ATP-dependent manner, leading to the release and dimerization of AriB.

Activated PARIS inhibits translation

We next investigated the consequences of PARIS activation by Ocr. The expression of Ocr from a plasmid in *E. coli* is sufficient to cause cell death in the presence of PARIS (Fig. 4a). Monitoring of PARIS⁺ cells under the microscope using live/dead staining with propidium iodide showed the

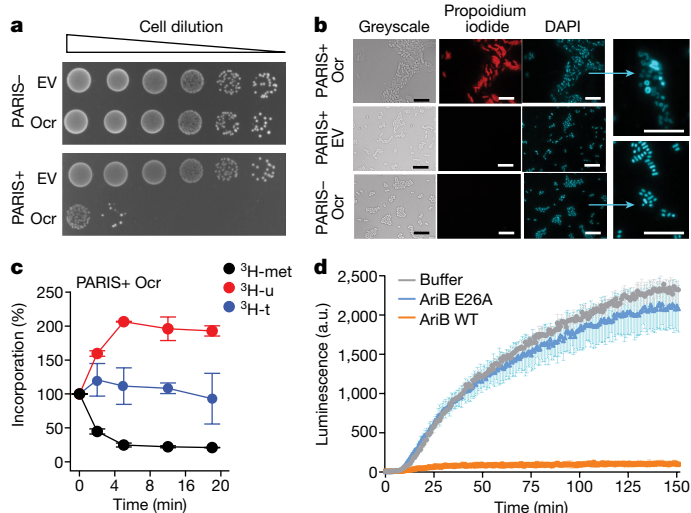


Fig. 4 | Activation of PARIS leads to cell death and inhibition of translation. **a**, Expression of Ocr from the pBAD vector in PARIS⁺ culture resulted in cell toxicity, whereas the empty vector (EV) had no impact. **b**, Ocr expression in PARIS⁺ cells induced cell death (propidium iodide staining, red) and DNA compaction (DAPI staining, blue). Arrows indicate zoomed-in images of nucleoid structures from PARIS⁺/Ocr (top) and PARIS⁻/Ocr (bottom). Cells were imaged 1 h following induction of Ocr expression. **c**, Metabolic labelling experiments show a drop in ³H-methionine incorporation following induction of Ocr, indicative of translation inhibition following PARIS activation. **d**, In vitro translation in the presence of activated AriB (WT) or active site mutant (E26A). Translation of firefly luciferase mRNA was monitored by luminescence. In the presence of activated AriB, luciferase signal was reduced markedly, further indicating that PARIS functions through a mechanism of translational inhibition. **c, d**, Mean and s.d. were calculated for each time point for three independent experiments. Scale bars, 10 μ m. a.u., Arbitrary units; 3H-t, 3H-thymidine.

accumulation of cells with permeabilized membrane within 20 min of Ocr induction (Fig. 4b and Supplementary Video 2). The TOPRIM domain, probably responsible for AriB toxicity, is frequently associated with DNase or RNase activities²⁹, but RNA or DNA preparations from AriB-activated cells did not show evidence of indiscriminate nuclease activity (Extended Data Fig. 8a–c). However, DAPI staining showed that PARIS activation results in nucleoid compaction (Fig. 4b), reminiscent of the translation inhibition phenotype³⁰. To understand the nature of this compaction, we performed chromosome conformation capture (Hi-C) experiments. The results showed a loss of chromosomal structures similar to that observed following treatment with chloramphenicol, a known inhibitor of translation (Extended Data Fig. 9).

To better understand the nature of PARIS-induced translational inhibition, we conducted metabolite-labelling experiments and in vitro translation assays. Production of Ocr in PARIS⁺ cells resulted in (1) rapid inhibition of ³H-methionine (³H-met) incorporation, consistent with translation inhibition (Fig. 4c) and (2) a moderate increase in ³H-uridine (³H-u) uptake. This increase is a known consequence of translation inhibition that can be explained by enhanced transcription of ribosomal RNA genes¹⁵. The importance of activated AriB in translational arrest was further corroborated using a luciferase reporter in an in vitro translation assay. The presence of activated AriB in the reaction mixture interferes with the production of luciferase, whereas the active site mutant (AriB E26A) has no impact (Fig. 4d). Collectively, these results demonstrate that PARIS activation results in translational inhibition.

Triggers and a suppressor of PARIS

Comparative analysis of related phages has provided considerable insight into both defence and antidefence strategies³¹. While testing

the activity of PARIS against phages of the T5 family, we noticed distinct plaquing phenotypes among T5 variants (Fig. 5a). PARIS does not protect cells from T5_{WT} phage infection, but efficiently blocks infection by a variant of T5 from the Moscow phage collection (T5_{Mos}). Consistent with translation arrest following PARIS activation, infection of PARIS⁺ cells with T5_{Mos} at high multiplicity of infection (MOI) resulted in rapid growth inhibition, which occurred more rapidly than the time required for lysis of PARIS⁻ cells (Extended Data Fig. 10a). Aside from a large deletion (Δ 30661–38625), the T5_{Mos} genome is nearly identical to the T5 archetype (Fig. 5b). We hypothesized that both phages contain a trigger that activates PARIS, but that the T5 archetype also contains a suppressor of PARIS that has been lost in T5_{Mos}. To determine how T5_{Mos} triggers PARIS, we isolated and sequenced T5_{Mos} mutants that escape PARIS immunity. All PARIS-escape mutants carried mutations in two genes encoding highly acidic proteins of unknown function: open reading frame (ORF)094 and ORF103 (ref. 32) (Fig. 5a–d). Expression of these proteins from a plasmid resulted in activation of PARIS toxicity (Fig. 5e), confirming that T5 carries two novel PARIS triggers. Mutated variants encoded by escape mutants did not trigger PARIS. We named these proteins PARIS triggers 1 and 2 (Ptr1, ORF094 and Ptr2, ORF103). These results show that structurally diverse, negatively charged viral proteins can activate the PARIS system. The presence of genes *ptr1* and *ptr2* in the genome of T5_{WT}, which is not affected by PARIS, further indicates that this phage contains a suppressor of PARIS immunity.

Viral tRNA is resistant to AriB cleavage

The T5 genome encodes 23 tRNAs, 16 of which have been lost in T5_{Mos}. Given the inhibitory effect of activated AriB on translation, we hypothesized that phage-encoded tRNAs are responsible for overcoming PARIS toxicity. To narrow the search for a PARIS suppressor, we tested a collection of T5 deletion mutants for their sensitivity to PARIS³³. Among the variants tested, T5₁₂₃ carries the smallest deletion in the tRNA genes region (Δ 29191–32442) and is still sensitive to PARIS (Extended Data Fig. 10b). The DNA fragment missing from T5₁₂₃ was divided into three segments, which were separately cloned on an expression plasmid and introduced into PARIS⁺ cells. A fragment encoding tRNA^{Pro}, tRNA^{fMet}, tRNA^{Lys} and tRNA^{Val} partially rescued the ability of phage T5₁₂₃ to infect PARIS⁺ cells, whereas other fragments had no effect (Extended Data Fig. 10c). Next, we cloned each of the tRNAs individually and tested their ability to rescue T5₁₂₃ infection. Overexpression of the T5 tRNA^{Lys} completely restored T5₁₂₃ infectivity (Fig. 5f), demonstrating that this phage-encoded tRNA neutralizes PARIS-induced toxicity and suggesting that the host tRNA^{Lys} is targeted by AriB.

To confirm that the *E. coli* tRNA^{Lys} is indeed degraded following PARIS activation, we performed RNA blot with total RNA extracted from PARIS⁺ cells 15 or 30 min following induction of the Ocr trigger. tRNA^{Lys} was degraded, whereas SYBR Gold staining did not show degradation of other RNAs (Fig. 5g). We note that we cannot exclude the possibility that other *E. coli* tRNAs might be specifically degraded by AriB, and that T5₁₂₃ complements their loss through some of the other tRNAs that it carries. More work is necessary to fully characterize the activity of AriB.

To determine how T5 tRNA^{Lys} suppresses PARIS immunity, we analysed RNA extracted from PARIS-activated cells expressing the phage tRNA from a plasmid. Although *E. coli* tRNA^{Lys} was still degraded, T5 tRNA^{Lys} remained intact (Fig. 5h and Extended Data Fig. 11a,b). The phage-encoded variant of tRNA^{Lys} contains mutations in the anticodon stem-loop (Extended Data Fig. 10d), which could be responsible for the lack of AriB recognition. To test this hypothesis, we introduced mutations from T5 tRNA into the *E. coli* tRNA^{Lys} counterpart. The resulting chimeric tRNA rescued infection by phage T5₁₂₃ in the presence of PARIS, confirming that AriB recognizes the tRNA anticodon stem-loop

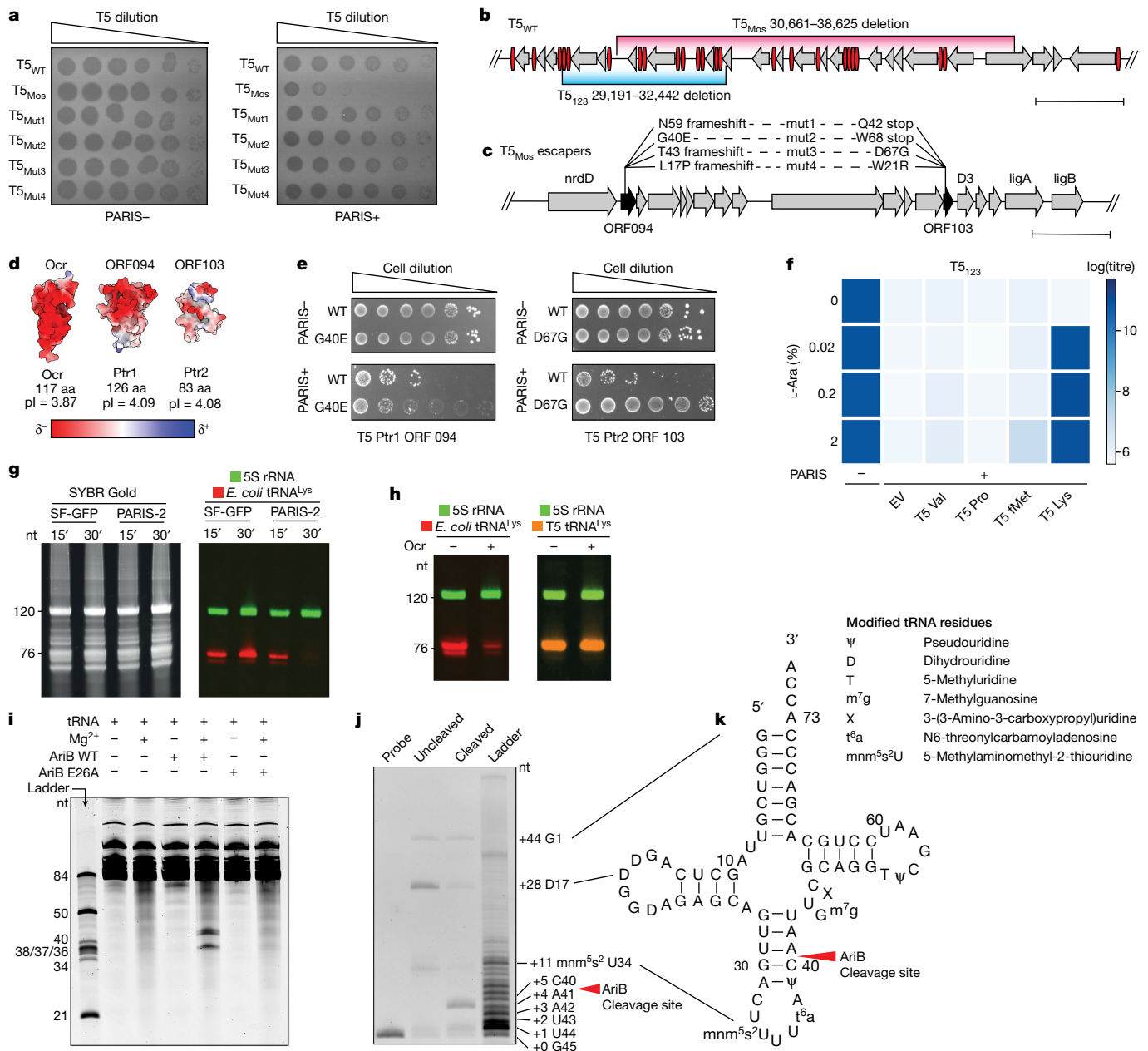


Fig. 5 | PARIS cleaves *E. coli* tRNA^{Lys} and T5 encodes a non-cleavable tRNA^{Lys}.

a, Plaque assays conducted using cells with (+) or without (–) PARIS against mutants T5_{WT}, T5_{Mos} and T5_{Mos} (T5_{Mut1–Mut4}). **b**, Graphical map of the WT T5 genome. ORFs are represented by grey arrows and tRNAs by red ovals. T5_{Mos} is missing an 8 kb fragment (red), with T5₁₂₃ having a smaller 3 kb deletion (blue). **c**, T5_{Mos} phages acquire mutations in ORF094 (Ptr1) and ORF103 (Ptr2) (black arrows) to escape PARIS-mediated immunity (mut1–mut4). **d**, AlphaFold2-predicted structures for Ptr1 and Ptr2 compared with T7 Ocr. Protein surfaces coloured by electrostatic potential. **e**, Expression of WT triggers, but not mutant, cause toxicity in PARIS⁺ cells but not in PARIS[–] cells. **f**, Overexpression of T5 tRNA^{Lys} in PARIS⁺ cells rescues infection by phage T5₁₂₃, which lacks tRNA^{Lys}. **g**, RNA blot performed on total RNA extracted from *E. coli* MG1655 carrying the PARIS system or a GFP control, and Ocr under the control of

a P_{Phf} promoter. RNA was extracted either 15 or 30 min (15' and 30', respectively) following induction. Identification of 5S rRNA (green) and *E. coli* tRNA^{Lys} (red) was made using distinct probes. **h**, Northern blot performed on total RNA extracted from *E. coli* carrying PARIS or GFP, Ocr and T5 tRNA^{Lys} under the control of an *araBAD* promoter. RNA was extracted following 30 min of induction. The tRNA^{Lys} (red) and T5 tRNA^{Lys} (yellow) probes show that the latter was not degraded during PARIS-mediated defence. **i**, Small RNA extracted from *E. coli* showing a cleavage product when incubated with AriB in a metal-dependent manner. **j**, Primer extension assay. AriB cleaves *E. coli* tRNA^{Lys} between positions 40 and 41. **k**, Schematic representation of the AriB cleavage site on *E. coli* tRNA^{Lys}. Scale bars, 2 kb. aa, Amino acids; L-ara, L-arabinose; pI, isoelectric point.

(Extended Data Fig. 10e). To determine the precise cleavage site of AriB, we incubated tRNAs extracted from *E. coli* with activated AriB, which resulted in cleavage products only in the presence of Mg²⁺ and with the catalytically active AriB (Fig. 5i). A primer extension assay performed using a probe complementary to the 3' end of the host tRNA^{Lys} showed that the AriB cleavage site lies between residues 40

and 41 of *E. coli* tRNA^{Lys} (Fig. 5j,k). This cleavage site was corroborated by Sanger sequencing of the cleaved 3' fragment (Extended Data Fig. 11c). Collectively, these results indicate that phage T5 encodes a non-cleavable variant of tRNA that carries a mutation in the AriB cleavage site, compensating for the loss of host tRNA depleted following PARIS activation.

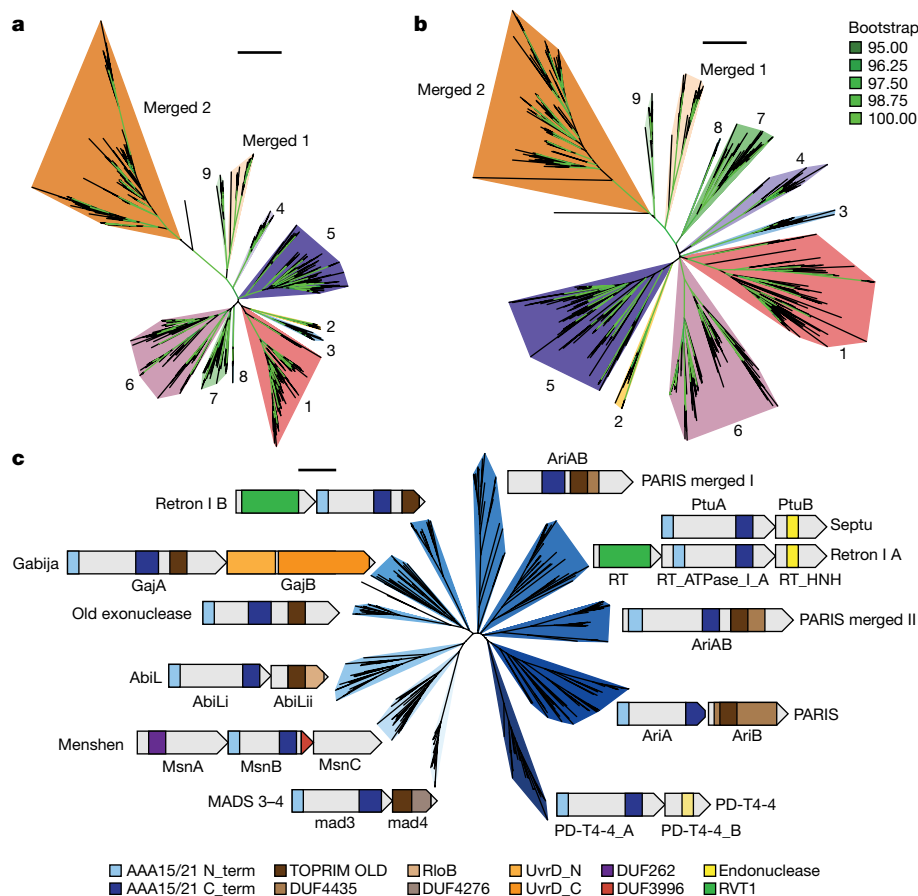


Fig. 6 | Phylogenies of PARIS- and related ABC ATPase-powered defence systems. a, b, The phylogenetic trees of AriA (a) and AriB (b). The two trees share similar branching patterns. Different colours represent 11 subclades of PARIS, including 2 in which *ariA* and *ariB* merged into a single *ariAB* gene (merged 1 and 2). **c,** Phylogenetic tree of AAA15/21 ATPase-containing defence

proteins. Pfam accessions AAA15 (PF13175), AAA21 (PF13304), TOPRIM-Old (PF20469), DUF4435 (PF14491), RloB (PF13707), DUF4276 (PF14103), UvrD_N (PF00580), UvrD_C (PF13361), DUF262 (PF03235), DUF3996 (PF13161), HNH endonuclease (PF14279, PF10107) and RVT1 (PF00078) were used for domain annotations. Tree scale bars, 1.

Evolution of the PARIS defence system

To inform our understanding of the diversity of PARIS systems and their relationship with other ABC ATPase-powered defence systems, we conducted phylogenetic analyses. First, we generated trees for AriA and AriB showing a diverse set of systems that can be grouped into 11 distinct clades, including 2 in which *ariA* and *ariB* merged into a single *ariAB* gene (Fig. 6a,b). *ariA* and *ariB* genes from the same system consistently fall in matching clades, showing that they coevolve and are not frequently swapped between systems. A rooted version of the AriB tree shows that AriAB emerged once from clade 9 of two gene systems in a single event and was then split into two different clades (Extended Data Fig. 11d,e). Interestingly, sequence alignments of AriA show that residues associated with the central and radial pores of the complex are poorly conserved (Extended Data Fig. 11d,e). This suggests that the signal transduction and effector parts of PARIS immunity are shared across homologues, whereas the central and radial pores have evolved to recognize different viral triggers.

PARIS was recently assigned to a family of OLD-like defence systems that share an ABC ATPase and a TOPRIM nuclease¹¹. This analysis identified four classes, in which the single-gene systems such as P2 Old are categorized as class 1, Gabija as class 2, reverse transcriptase-containing systems as class 3 and PARIS systems as class 4. We built a tree of known antiphage defence systems that carry an ABC ATPase of the AAA15/21 family (Fig. 6c). Our analysis adds AbiL and MADS3–4 systems to the list of systems sharing an ABC ATPase and TOPRIM nuclease; it also shows how this ABC ATPase can be found in association with a variety

of other effector domains, demonstrating the evolutionary success of this domain as the probable sensor of viral infection in diverse immune systems.

Discussion

Our results demonstrate how AriA forms a propeller-shaped hexamer capable of binding three AriB monomers. AriA is an ATP-dependent sensor that detects viral immune suppressors (for example, Ocr), which trigger the release of AriB. Released AriB assembles into a homodimer that cleaves specific host tRNAs, resulting in rapid interruption of translation followed by loss of membrane integrity. As opposed to prototypical toxin–antitoxin systems, AriB has modest toxicity when expressed alone. This suggests that AriA activates AriB through a structural modification that enables dimerization or an unidentified post-translational modification. More work will be necessary to determine the mechanism of AriB release and activation.

Interestingly, we found that the PARIS system from *E. coli* B185 is triggered not only by the phage T7 Ocr protein but also by the Ptr1 and Ptr2 proteins of phage T5. These proteins are small and negatively charged but lack obvious sequence or structural similarities. Determination of how PARIS detects such diverse proteins, and whether different systems have distinct trigger sensitivities, presents an intriguing avenue for future research.

Our work adds to a growing list of defence systems that cause translational arrest following infection. The PrrC and Retron I A (that is, PtuAB proteins) systems induce tRNA degradation following detection

of foreign proteins, whereas Cas13a degrades tRNA following detection of foreign RNA^{34–36}. However, translational arrest is not limited to tRNA degradation, Mogila et al.³⁷ recently identified a CRISPR–Cas system that cleaves mRNA associated with the ribosome, which arrests translation, and the protease late inhibition of T4 (Lit) degrades the translation elongation factor EF-Tu following detection of the major capsid protein of phage T4 (ref. 38).

We show that some phages circumvent PARIS-induced toxicity by expression of non-cleavable tRNA variants. Many tailed phages, such as those from the families Demerecviridae (T5-like phages), Straboviridae (T4-like phages) or Ackermannviridae, encode large arrays of tRNAs previously thought to be associated with translation optimization due to codon bias in the host and viral genome³⁹. However, estimation of the abundance of host and phage tRNAs during infection, compared with the corresponding codon frequencies in produced mRNAs, does not directly support this hypothesis³⁹. A recent computational analysis suggests that phages may have evolved these variants in response to bacterial tRNA-targeting toxins, as evidenced by distinct anticodon loop structures aimed at evasion of host nucleases⁴⁰. This hypothesis is further supported by a parallel work demonstrating that phage T5 tRNA^{Tyr} provides protection against the PtuAB toxin of the Eco7 (Ec78) retron³⁶.

Interestingly, this strategy of encoding tRNA clusters is also observed in some eukaryotic viruses, such as those from the families Herpesviridae, Mimiviridae and Phycodnaviridae⁴¹, possibly serving a similar function in evasion of host immunity. Strikingly, an anticodon nuclease was also recently found to protect humans against Pox viruses, demonstrating that the inhibition of translation through the inactivation of specific tRNAs is a conserved antiviral strategy used across domains of life⁴².

Online content

Any methods, additional references, Nature Portfolio reporting summaries, source data, extended data, supplementary information, acknowledgements, peer review information; details of author contributions and competing interests; and statements of data and code availability are available at <https://doi.org/10.1038/s41586-024-07874-3>.

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Methods

Bacterial strains, phages and plasmids

The phages, bacterial strains and plasmids used in the study are listed in Supplementary Table 1 and the primers are listed in Supplementary Table 2. Infection with T5 phage was carried out in Luria-Bertani (LB) medium supplemented with 1 mM CaCl₂. Unless otherwise indicated, 0.2% l-ara, 0.1 mM isopropyl-β-D-thiogalactopyranoside (IPTG) and 0.2 μg ml⁻¹ anhydrotetracycline (aTc) were used for induction.

For structural studies, the coding sequences for AriA and AriB from *E. coli* B185 (ref. 2) were cloned under the control of a T7 promoter into a pRSF-Duet vector with a C-terminal strep-tag II on AriB. For coexpression of PARIS with the T7 Ocr trigger, the TOPRIM nuclease of AriB was inactivated (E26A) and strep-tag II was removed by site-directed mutagenesis. The coding sequence for T7 Ocr protein was cloned into a pET-Duet vector with a C-terminal strep-tag II. For in vitro binding assays, the coding sequence for T7 Ocr protein was cloned into a pET-Duet vector with an N-terminal His6-TwinStrep-SUMO tag. All in vivo experiments were performed with pFR85, encoding PARIS from *E. coli* B185 with its native promoter and an inducible *P_{tet}* promoter upstream. For in vivo protein pull-downs, C-terminal AriB strep-tag II was introduced to pFR85. Mutants of PARIS were constructed from pFR85 using either the Gibson method⁴³ or Q5 site-directed mutagenesis kit (NEB). PARIS triggers both Ptr1 (T5 ORF094) and Ptr2 (T5 ORF103) from the T5 phage, and T5 genome fragments, T5 tRNA^{Lys} and *E. coli* tRNA^{Lys} were cloned on the pBAD18 vector under the control of the *araBAD* promoter using the Gibson method. Twenty base pairs upstream and downstream of tRNA, genes were preserved during cloning. T5 nucleotide positions are provided according to genome assembly AY543070.1. All constructions were verified using Sanger sequencing.

Expression and purification of the PARIS complex

AriA and AriB-Strep were coexpressed from a pRSF-Duet plasmid in BL21-AI cells (Invitrogen). Cultures grown at 37 °C to 0.4–0.5 OD₆₀₀ were induced with 0.1 mM IPTG and 0.1% l-ara. Following induction, cultures were incubated overnight at 16 °C. Cells were pelleted (3,000g, 10 min at 4 °C), resuspended in lysis buffer (25 mM Tris pH 8.5, 150 mM NaCl and 1 mM EDTA) and lysed by sonication. Cellular debris was removed by centrifugation (10,000g, 25 min at 4 °C) and the AriAB complex was purified by affinity chromatography on 5 ml of streptactin resin (IBA) and eluted with 2.5 mM desthiobiotin (IBA) in lysis buffer. The eluted protein was concentrated (100K MWCO PES Spin-X UF concentrator, Corning) and further purified by size exclusion chromatography (Sup200 column, Cytiva) in lysis buffer with 2% glycerol. Fractions of interest were combined and concentrated to 5 μM (100K MWCO PES concentrator, Pierce) and used immediately for vitrification on cryo-EM grids.

In vivo protein pull-down

For determination of protein-of-interest interacting partners in vivo, either AriB-Strep expressed from pFR85 or Ocr-Strep expressed from pBAD was used as bait, and pull-downs were performed in *E. coli* BW25113. Three litres of cell culture were grown in LB medium at 37 °C in Thomson flasks with aerated lids (1.5 litres per flask). Expression of proteins was induced with either 0.2 μg ml⁻¹ aTc or 0.2% l-ara at an approximate OD₆₀₀ of 0.1, and cells were harvested by centrifugation after reaching an approximate OD₆₀₀ of 0.8–1.0. Cells were processed as described above, and Strep-tagged proteins purified on two stacked 5 ml StrepTrap HP (Cytiva) columns. The identity of protein bands following SDS–polyacrylamide gel electrophoresis (PAGE) was determined by matrix-assisted laser desorption/ionization time-of-flight mass spectrometry. Roughly one-third of the Coomassie-stained band was cleaved from the gel, and samples were prepared with Trypsin Gold (Promega) in accordance with the manufacturer's instructions. Mass spectra were obtained using the rapifleX system (Bruker).

For the production of activated AriB, AriB-Strep from pFR85 and non-tagged Ocr from pBAD were expressed in separate cultures and cells grown as previously described. Cell cultures were mixed, and cell walls disrupted by sonication on ice (60% power, 10 s pulse, 20 s pause, 30–60 min) on a Qsonica sonicator with 6 mm sonotrode, followed by StrepTrap HP purification of activated AriB-Strep. The molecular weight of protein complexes was determined following size exclusion chromatography performed on a Superdex 200 Increase 10/300 column (GE Healthcare) calibrated with the High Molecular Weight calibration kit (GE Healthcare).

Glutaraldehyde cross-linking assay

To determine the oligomeric state of activated AriB in solution, we performed a glutaraldehyde cross-linking assay. Concentrated AriB aliquots were mixed with glutaraldehyde (no. 253857.1611, PanReac) to a final concentration of 0.025, 0.125 or 0.25% v/v and incubated for 30 min at room temperature. The reaction was quenched by the addition of one sample volume of 1 M Tris-HCl at pH 8.0 and a half-sample volume of 5× concentrated SDS sample buffer. Samples were heated for 5 min at 60 °C and separated by SDS–PAGE.

In vitro protein pull-down

The T7 Ocr protein with an N-terminal His6-TwinStrep-SUMO was expressed from a pET-Duet plasmid in BL21(DE3) cells. Cultures grown at 37 °C to 0.4–0.5 OD₆₀₀ were induced with 0.5 mM IPTG. Following induction, cultures were incubated overnight at 16 °C. Cells were pelleted (3,000g, 10 min at 4 °C), resuspended in lysis buffer (25 mM Tris pH 8.5, 150 mM NaCl and 1 mM EDTA) and lysed by sonication. Cellular debris was removed by centrifugation (10,000g, 25 min at 4 °C) and HSS-Ocr was purified by affinity chromatography on 5 ml of streptactin resin (IBA) and eluted with 2.5 mM desthiobiotin (IBA) in lysis buffer. The eluted protein was incubated with SUMO protease at 4 °C overnight before washing over 5 ml of Ni-NTA resin (25 mM Tris pH 7.5, 300 mM NaCl, 1 mM TCEP and 20 mM Imidazole) to remove the SUMO-cleaved tag. Flow-through protein was concentrated (10K MWCO PES Spin-X UF concentrator, Corning) and further purified by size exclusion chromatography (Sup6 10 300 column, Cytiva) in lysis buffer with 2% glycerol. Fractions of interest were combined and concentrated (10K MWCO PES Spin-X UF concentrator, Corning). Cleaved Ocr (3.8 μM) was incubated with WT PARIS (11.3 μM) at 37 °C for 15 min before analysis via size exclusion chromatography (Sup6 10 300 column, Cytiva) in lysis buffer with 2% glycerol. Fractions of interest were combined, concentrated (10K MWCO PES Spin-X UF concentrator, Corning) and analysed via SDS–PAGE (15% acrylamide, 150 V for about 1 h).

Cryo-EM sample preparation and data collection

All samples were subjected to vitrification on 1.2/1.3 Carbon Quantifoil grids, which were glow-discharged using an easiGlow (Pelco) glow-discharge station (glow 45 s, hold 15 s). Following the application of 3 μl of 5 μM PARIS + 1 mM ATPγS, grids were subjected to double-sided blotting using a VitroBot Mk IV, with a force of 5 for 4 s at 4 °C and 100% relative humidity. Following plunge-freezing in liquid ethane, grids were clipped and stored in liquid nitrogen.

For PARIS obtained from AriB pull-downs, cryo-EM grids were screened and all data collected using a Talos Arctica microscope (ThermoFisher) operating at 200 kV and equipped with a K3 direct electron detector (Gatan). SerialEM⁴⁴ software v.3.8.6 was used to automate data collection. For final data collection, a total of 7,340 movies were collected in super-resolution mode with a pixel size of 0.552 Å and a total dose of 56.42 e⁻Å⁻² distributed over 50 subframes.

For determination of the structure of AriA, samples were prepared for cryo-EM analysis as described above. Following screening of grids, 10,340 movies were collected using a Talos Arctica microscope (ThermoFisher) operating at 200 kV and equipped with a K3 direct electron detector (Gatan). SerialEM software v.3.8.6 was used to automate data

collection. Videos were recorded in super-resolution mode with a pixel size of 0.552 Å and a total dose of 56.31 eÅ⁻² distributed over 50 sub-frames.

Cryo-EM data processing of inactive PARIS

All data were processed in cryoSPARC v.4.41 (ref. 45). Following patch motion correction and contrast transfer function (CTF) estimation, micrographs with CTF fits below 8 Å were discarded, yielding 5,998 movies for particle picking. Following processing of a subset of 200 micrographs, the AriA₆-AriB₃ assembly of the PARIS complex was apparent. Using this low-resolution initial volume, AlphaFold2-predicted structures of AriA and AriB were fit into the map and used to generate a 20 Å low-pass-filtered map using the Molmap command in ChimeraX v.1.7 (ref. 46). This volume was imported to cryoSPARC and used to generate templates for particle picking. A total of 4,078,384 particles were extracted and, following initial two-dimensional classification, 1,643,515 were retained for further processing. These particles were sorted using a three-class *ab initio* reconstruction that showed two junk classes and a volume corresponding to the assembled PARIS complex containing 940,782 particles. After a second round of two-dimensional classification, 934,763 particles remained and were subjected to a further round of *ab initio* reconstruction and heterogeneous refinement. This produced a consensus volume containing 532,010 particles with conformational heterogeneity for two AriB subunits, with one asymmetric unit of the complex being well aligned. Masked local refinement was used to determine a 3.2-Å-resolution map of one asymmetric unit of the complex. The consensus refinement with 532,010 particles was also subjected to multiple rounds of masked three-dimensional classification to isolate particles in the *cis* and *trans* arrangements. The fully assembled PARIS complex demonstrates flexibility based on three-dimensional variability analysis, which limited the attainable resolution of the fully assembled complex. The density maps for both local refinement and the full complex are available at EMDB-42719, 43103 and 43104, and raw micrographs are available at EMPIAR-11832.

Cryo-EM data processing of Ocr pulldown

All data were processed in cryoSPARC v.4.41. Of the 10,340 movies collected, 9,399 were selected for further processing based on the CTF fit criteria described above. Using a blob picker with a particle diameter of 200 Å, particles were extracted and subjected to two-dimensional classification. In the two-dimensional classes there were two distinct macromolecular complexes present, one of which was interpreted to be the AriA scaffold of the PARIS complex, with the other interpreted as corresponding to RNA polymerase.

From 7 selected classes corresponding to the AriA hexamer, 667,479 particles were selected for downstream refinement. From a two-class *ab initio* reconstruction of these particles, 392,989 were retained. Iterative sorting by multiclass *ab initio* reconstruction and heterogeneous refinements yielded a volume containing 62,732 particles. AriA subunits can be fit into this density, but poor resolution rendered this structure indistinguishable from the AriA scaffold identified in the intact PARIS complex. No density was observed for Ocr, which suggests that multiple binding sites may be present on the PARIS complex, or that Ocr is attached to AriA in a flexible state. The density map is available at EMDB-43105 and raw micrographs at EMPIAR-11833.

Model building and refinement

To determine the structure of the asymmetric unit of the PARIS complex, density half-maps were obtained from cryoSPARC local refinements before being provided as inputs to DeepEMhancer v.0.14 (ref. 47) for map sharpening. AlphaFold2-predicted structures of AriA and AriB were then docked into the map density, and map-to-model fit was improved using the ISOLDE v.1.7.1 (ref. 48) molecular dynamics simulation environment in ChimeraX v.1.7. Following initial equilibration, PDB files were saved and used as inputs for real-space refinement using

PHENIX v.1.20.1 (ref. 49). Following initial refinement, problematic areas were manually inspected and modified using COOT v.0.9.8.1 (ref. 50), and side-chains were removed from regions with resolution below 4 Å. Following addressing of clashes and non-rotameric side-chain orientations, iterative refinements were carried out in Phenix until Clashscores, Ramachandran and side-chain outliers failed to improve. The final model and corresponding map have been deposited (PDB ID: 8UX9, EMDB-42719).

To generate the biological assemblies for the PARIS complex atomic model, half-maps from C3 symmetric and non-uniform refinements of particles corresponding to the *cis* and *trans* orientations were obtained from cryoSPARC and sharpened in DeepEMhancer v.0.14. Following fitting of the three copies of the asymmetric unit into the *cis* and *trans* density maps using the ChimeraX v.1.7 fitmap command, the find NCS tool in Phenix was then used to calculate the appropriate symmetry operators for the assembled forms of the complex. These symmetry operators were applied to the asymmetric unit to generate biological assemblies 1 and 2 using the Apply NCS tool in Phenix. Maps for the *cis* and *trans* orientations of the PARIS complex are available at EMDB-43104 and EMDB-43105, respectively.

ATPase assays

ATPase activity was measured using 6 μM Ocr, 100 nM PARIS and 128 μM ATP supplemented with 10 nM [α-32 P]-ATP (PerkinElmer). The reaction buffer included 20 mM Tris-HCl pH 8.0, 150 mM sodium chloride, 1 mM DTT and 5 mM magnesium chloride. Reactions were incubated at 37 °C, and 10 μl aliquots were quenched with phenol at 1, 2, 4, 8, 16 and 32 min. For trigger-only reactions, 100 nM Ocr was mixed with 128 μM ATP supplemented with 10 nM [α-32 P]-ATP (PerkinElmer) in the reaction buffer. Reactions were incubated for 32 min at 37 °C and quenched with phenol. The ATP and ADP makers were created using T4 polynucleotide kinase (NEB). T4 PNK was mixed with 128 μM ATP, supplemented with 10 nM [α-32 P]-ATP and 1 mM DNA oligo, in T4 polynucleotide kinase buffer (NEB); this mixture was then incubated at 37 °C for 1 h. All reaction products were phenol-chloroform extracted and resolved on silica TLC plates (Millipore). Reaction products were spotted 2.5 cm above the bottom of the TLC plate. Plates were placed in a TLC developing chamber filled to around 1.5 cm with developing solvent (0.2 M ammonium bicarbonate pH 9.3, 70% ethanol and 30% water) and covered with aluminium foil for 4 h at room temperature. Plates were then exposed to a phosphor screen, imaged using a Typhoon phosphor imager and quantified with the ImageQuant software package (Cytiva).

EOP assay

The activity of PARIS mutants was measured in comparison with the WT system by performing efficiency of plaquing (EOP) assays with phage T7. *E. coli* K-12 MG1655 carrying each of the AriA or AriB mutants, the control plasmid pFR66 (sfGFP) or the WT PARIS system (pFR85) was grown overnight in LB + kanamycin (50 μg ml⁻¹). Bacterial lawns were prepared by mixing 100 μl of a stationary culture with 5 ml of LB + 0.5% agar, and the mixture was poured onto a Petri dish containing LB + kanamycin (50 μg ml⁻¹). Tenfold serial dilutions of high-titre stock of T7 phage were spotted on each plate and incubated for 5 h at 37 °C. EOPs with T5 and T5-like phages were performed with *E. coli* BW25113 as described above, except that plates were incubated at 37 °C overnight and the top agar was supplemented with 1 mM CaCl₂. Induction of tRNA genes or phage T5 fragments was achieved by supplementing the top agar with the indicated amount of l-ara. Plaque assays were performed in at least two independent replicates.

Liquid culture phage infection

To monitor the dynamics of T5_{Mos} phage infection of PARIS⁺ (pFR85) and PARIS⁻ (pFR66) cultures in liquid, we used an EnSpire Multimode Plate Reader (PerkinElmer). Overnight bacterial cultures were diluted

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100-fold in LB medium with appropriate antibiotics and grown at 37 °C in 10 ml of LB supplemented with 1 mM CaCl₂. At OD₆₀₀ 0.6, 200 µl aliquots were transferred to 96-well plates and infected at the indicated MOI. Optical density was monitored for 10 h. All experiments were performed in three biological replicates.

Solid medium toxicity assay

PARIS toxicity in the presence of T7 Ocr or Ptr1 and Ptr2 was measured using a spot-test assay. PARIS⁺ (pFR85) and PARIS⁻ (pFR66) cultures carrying the pBAD vector-encoding-indicated trigger were grown overnight in 10 ml of LB at 37 °C. Stationary cultures were diluted to OD₆₀₀ 0.6 and plated on LB or minimal-medium (M9 with 5% v/v LB) agar plates supplemented with 0.2% L-ara by serial tenfold dilution. Control plates without induction contained 0.2% glucose to prevent leakage of the *araBAD* promoter. To assess the toxicity of the PARIS AriA D401N/E404Q mutant, we used either pFR85 (PARIS expression driven from native promoter, BW25113 *E. coli* strain) or pRAW 464 vector (PARIS overexpression from T7 promoter, BL-21 (DE3) *E. coli* strain). The cultures were plated on M9 as described above, and PARIS expression was induced with 1 mM IPTG. Ocr expression was driven from pBAD with 0.2% L-ara.

Liquid culture toxicity assay

To assess the toxicity of AriB expression alone compared with activated PARIS, *E. coli* BL-21 AI cells were transformed with plasmids encoding PARIS (pRAW 464) and AriB (pRAW 466) or PARIS (pRAW 464) and Ocr (pRAW 472). Following overnight growth of cultures in LB supplemented with the appropriate antibiotic, cells were diluted to OD₆₀₀ 0.1 in LB + antibiotics before growing to OD₆₀₀ 0.5. After cells had reached OD₆₀₀ 0.5, 200 µl of culture was added to 800 µl of medium supplemented with antibiotics and either 0.4% glucose, 0.2% L-ara and 0.1 mM IPTG or 0.5% L-ara and 0.1 mM IPTG. Next, 200 µl of each culture was aliquoted into a 96-well plate (Evergreen, no. 222-8030-01F). Growth was monitored using an Agilent BioTek Cytation5 plate reader set to 30 °C with continuous linear shaking, and OD₆₀₀ readings taken every 5 min for 15 h.

Escherichia coli MG1655 strains carrying plasmids pFR85 (P_{tet} ariAB), pFR85ΔAriA (pFR99), pFR85ΔAriB (pFR100) or pFR85-AriA(R212*) (pFD294), in which a stop codon was introduced in AriA, truncating the protein before the domain that interacts with AriB, were diluted 1:100 from an overnight culture in LB + kanamycin (50 µg ml⁻¹) and arrayed in a 96-well plate. Growth was then monitored every 10 min for 30 h at OD₆₀₀ in an infinite M200 Pro plate reader (Tecan) at 37 °C with shaking. Absorbance values of three biological replicates were analysed using GraphPad Prism v.10.1.2. Standard two-way analysis of variance was used to calculate *P* values for growth rates relative to WT PARIS.

Fluorescence microscopy

PARIS⁺ (pFR85) and PARIS⁻ (pFR66) carrying pBAD Ocr were diluted 1:100 and cultivated in LB supplemented with appropriate antibiotics at 37 °C. When OD₆₀₀ reached 0.4, an aliquot was mixed with DAPI (Invitrogen) at 1 mg ml⁻¹ final concentration and propidium iodide (Invitrogen) at 1 µg ml⁻¹ final concentration and incubated at room temperature in the dark for 5 min. LB + 1.5% agarose slabs (approximately 0.2 mm thick) were prepared on a 75 × 25 mm² microscopy slide (Fisher Scientific). Agarose slabs contained staining dyes at the same concentration were optionally supplemented with 0.2% L-ara, to induce Ocr expression. Roughly 1 µl of stained cells was placed on an agarose slab and, following 1 min of drying, the slab was covered by a small (22 × 22 mm²) coverslip (Fisher Scientific). Imaging was performed on a Nikon Eclipse Ti-E inverted epifluorescence microscope. Each field of view was imaged in the transmitted light channel (200 ms exposure) and in the DAPI (filterset Semrock DAPI-50LP-A, 200 ms exposure) and propidium iodide (filterset TxRed-4040C, 200 ms exposure) channels.

Extraction of total DNA

Total DNA was extracted at different time points (15, 30, 60 and 120 min) with or without DAPG to induce the expression of Ocr in the presence of PARIS. DNA was extracted using the Wizard genomic DNA purification kit (Promega) according to the manufacturer's instructions and treated with RNase at 0.1 mg ml⁻¹. Each sample of total DNA was loaded on 1% agarose 1× TAE gel.

TUNEL assay

TUNEL assays was performed to estimate in vivo the accumulation of dsDNA breaks, according to methods described previously⁵¹. PARIS⁺ (pFR85) culture carrying the pBAD Ocr vector was grown to OD₆₀₀ 0.3, followed by Ocr induction with 0.2% L-ara for 2 h. As a positive control, cells were treated with 0.1% H₂O₂ to induce accumulation of DNA breaks, then 2 ml of cells was harvested by centrifugation and treated according to the standard protocol (TUNEL Assay Kit – FITC, no. ab66108, abcam). Measurement of fluorescein isothiocyanate fluorescence was performed with the CytoFLEX cytometer (Beckman Coulter) in the fluorescein isothiocyanate channel. A total of 100,000 events were collected for each sample in three biological replicates. Data were analysed and visualized in FlowJo v.10 (ref. 52).

Hi-C procedure and sequencing

Escherichia coli MG1655 carrying plasmids pFR85 (P_{tet} ariAB) and pFD250 (P_{Phf} ocr), or control plasmids pFR66 (sfGFP) and pFD245 (PhIF sfGFP), was diluted 1:100 from an overnight culture in LB + kanamycin (50 µg ml⁻¹) + chloramphenicol (20 µg ml⁻¹) and grown to OD₆₀₀ 0.3. DAPG 50 µM was then added, followed by incubation for 15 and 30 min. To compare the effect of PARIS activation with that of treatment with chloramphenicol, *E. coli* MG1655 harbouring pFR66 was diluted 1:100 from an overnight culture in LB + kanamycin (50 µg ml⁻¹) to OD₆₀₀ 0.3, followed by the addition of chloramphenicol (20 µg ml⁻¹) and incubation for 15 and 30 min. Cell fixation was performed with 4% formaldehyde (Sigma-Aldrich, catalogue no. F8775) as described in ref. 53. Quenching of formaldehyde with 300 mM glycine was performed at 4 °C for 20 min. Hi-C experiments were performed with the Arima kit. Samples were sonicated using Covaris (DNA 300 base pairs).

Preparation of samples for paired-end sequencing was performed using the Invitrogen TM Colibri TM PS DNA Library Prep Kit for Illumina according to the manufacturer's instructions; the detailed protocol is available in ref. 53. Reads were aligned with bowtie2 v.2.4.4, and Hi-C contact maps were generated using hicstuff v.3.0.3 (<https://github.com/koszullab/hicstuff>) with default parameters and using the HpaII enzyme for digestion. Contacts were filtered as described in ref. 54, and PCR duplicates (defined as paired reads mapping at exactly the same position) were discarded. Matrices were binned at 4 kb. Balanced normalizations were performed using the ICE algorithm⁵⁵. For all comparative analyses, matrices were downsampled to the same number of contacts. The Hi-C signal was computed as contacts between adjacent 5 kb bins, as described in ref. 56. For comparison of this signal with other genomics tracks, we binned it at the desired resolution.

Metabolic labelling

Metabolic labelling with 3H-met, 3H-u and 3H-t in experiments was performed as described previously⁵⁷, with minor modifications. *E. coli* BW25113 PARIS⁺ (pFR85) strain was transformed with the pBAD plasmid carrying the Ocr trigger (for L-ara-inducible expression). Transformed cells were initially plated on LB plates supplemented with 100 µg ml⁻¹ ampicillin, 25 µg ml⁻¹ kanamycin and 0.2% glucose. Using individual *E. coli* colonies for inoculation, 2 ml liquid cultures were prepared in defined Neidhardt MOPS minimal medium (supplemented with 100 µg ml⁻¹ ampicillin, 25 µg ml⁻¹ kanamycin, 0.1% casamino acids and 1% glucose) and grown overnight at 37 °C with shaking. Subsequently, experimental 20 ml cultures were prepared in 125 ml conical flasks in

MOPS medium, supplemented with 0.5% glycerol, 100 $\mu\text{g ml}^{-1}$ ampicillin and 25 $\mu\text{g ml}^{-1}$ kanamycin, as well as a set of 19 amino acids (lacking methionine), each at a final concentration of 25 $\mu\text{g ml}^{-1}$. These cultures were inoculated at OD_{600} 0.05 and grown at 37 °C with shaking to OD_{600} 0.3. At this point, 1 ml aliquots (designated as the preinduction zero time point) were transferred to 1.5 ml Eppendorf tubes containing 10 μl of the respective radioisotope (3H-met (0.77 μCi , PerkinElmer), 3H-u (0.1 μCi , PerkinElmer) or 3H-t (0.32 μCi , PerkinElmer)) preincubated at 37 °C. Concurrently, *Ocr* expression in the remaining 19 ml culture was induced by the addition of *L*-ara to a final concentration of 0.2%. Throughout the *Ocr* induction time course, 1 ml aliquots were taken from the 20 ml culture and transferred to 1.5 ml Eppendorf tubes containing 10 μl of the appropriate radioisotope. Radioisotope incorporation was halted after 8 min of incubation at 37 °C by the addition of 200 μl of ice-cold 50% trichloroacetic acid (TCA) to the 1 ml cultures. In addition, 1 ml aliquots were sampled for OD_{600} measurement during the induction time course. The resultant 1.2 ml TCA-halted culture samples were loaded onto GF/C filters (Whatman) prewashed with 5% TCA, and the unincorporated label was removed by washing the filter twice with 5 ml of ice-cold TCA followed by two 5 ml washes with 95% EtOH. The filters were placed in scintillation vials and dried for 2 h at room temperature, followed by the addition of 5 ml of EcoLite-scintillation cocktail (MP Biomedicals). Following shaking for 15 min, radioactivity was quantified using the automatic TDCR Liquid Scintillation Counter (HIDEX). Isotope incorporation was quantified by normalization of radioactivity counts (CPM) to OD_{600} , with the preinduction zero time point serving as the reference (set to 100%). All experiments were performed as biological triplicates using three independent liquid starter cultures inoculated with different colonies.

In vitro translation

In vitro translation was monitored by the production of luciferase signal in a PURExpress in vitro protein synthesis kit (NEB), using firefly luciferase mRNA as an input. pT7-Fluc, encoding the firefly luciferase (*fluc*) gene under the control of the T7 promoter, was used as a template for T7 in vitro transcription with the MEGascript kit (Thermo Scientific). mRNA was treated with DNase I (Thermo Scientific) and purified with the Monarch PCR & DNA cleanup kit (NEB). The PURExpress in vitro translation reaction was assembled in RNase-free tubes according to the manufacturer's instructions, with minor modifications. Reaction mixtures were adjusted to a total volume of 5 μl (2 μl of solution A, 1.5 μl of solution B, 0.3 μl of RiboLock (40 U μl^{-1} , Thermo Scientific) and 0.2 μl of D-luciferin (10 mM, Sigma)), supplemented with either 0.5 μl of activated AriB or AriB (E26A) (100 $\mu\text{g ml}^{-1}$ final concentration) or the same volume of buffer A (50 mM NaCl, 1 mM EDTA, 100 mM Tris-HCl and 5 mM β -ME, pH 8.0) as a control. The reaction was started by the addition of 0.5 μl of purified *fluc* mRNA (500 ng), immediately placed in a 384-well white plate and covered with optically clear film. Accumulation of luminescent signal was measured with the EnSpire Multimode Plate Reader (PerkinElmer) for 3 h at 37 °C. AriB concentration was determined with the Qubit Protein Broad Range Assay Kit (Invitrogen).

Isolation and sequencing of T5 escaper mutants

For the selection of PARIS-escape mutants, phage T5_{Mos} was continuously incubated with PARIS⁺ (pFR85) culture for 3 days. In brief, after reaching OD_{600} 0.6, cells were infected with T5_{Mos} at an approximate MOI of 0.1 and the culture incubated overnight at 37 °C. Phage was collected the following day and 1 ml of lysate was used to initiate the next round of infection with fresh PARIS⁺ (pFR85) culture. Escaper mutants were purified from single plaques obtained from PARIS⁺ (pFR85) culture and produced on the PARIS⁻ (pFR66) culture. Eight millilitres of high-titre lysate (around 10^{10} plaque-forming units per millilitre) was PEG precipitated, and phage genomic DNA was purified as described previously⁵⁸. DNA libraries were prepared according to a

standard protocol and sequenced on the MiniSeq platform (Illumina) with paired-end 150 cycles (75 + 75). Genome assemblies were performed with SPAdes implemented in Unicycler⁵⁹. To identify mutations, genomes of T5_{Mos} escapers were aligned with the sequenced genomes of T5_{Mos} and T5_{wt} from initial stocks. T5 strains reported by Glukhov et al.³³ were sequenced on the DNBSeg-G400 platform (BGI) to validate boundaries of deletions.

Extraction of total RNA

Total RNA was isolated from *E. coli* strain MG1655 harbouring either pFR85 (P_{tet} *ariAB*) or control plasmid pFR66 (P_{tet} *sGFP*). Cells also carried pFD250 (P_{PhIF} *ocr*) with *Ocr* under the control of an inducible DAPG P_{PhIF} promoter (Fig. 5g and Extended Data Fig. 8b), as well as plasmid pFD287 carrying the tRNA^{Lys} of phage T5 under the control of the *araBAD* promoter inducible by *L*-ara (Fig. 5h).

Cells were diluted 1:100 in LB with the appropriate antibiotics from overnight cultures and grown with agitation at 37 °C. When cultures reached OD_{600} 0.2, aTc was added to a final concentration of 0.5 $\mu\text{g ml}^{-1}$ to induce expression of the PARIS system. For the experiment shown in Fig 5h, *L*-ara was added to a final concentration of 0.2% to induce T5 tRNA^{Lys} from plasmid pFD287. Growth continued to OD_{600} 0.4, and the *Ocr* trigger was induced with DAPG added to a final concentration of 50 μM , followed by incubation for an additional 15 and 30 min post induction.

Cells were centrifuged at 4,000g for 10 min, resuspended with 0.2 ml of lysozyme buffer (20 mM Tris-HCl pH 8.0, 2 mM EDTA, 1% Triton X-100 and 20 mg ml^{-1} lysozyme) and lysed for 30 min at 37 °C. One millilitre of TRIzol reagent (Zymo Research) was added to the samples, with incubation for 5 min at room temperature. Lysates were extracted with 0.2 ml of cold chloroform and centrifuged at 12,000g for 15 min at 4 °C. The aqueous phase of the sample was precipitated with 0.5 ml of 100% cold isopropanol, incubated for 10 min at room temperature and centrifuged at 12,000g for 10 min at 4 °C. RNA pellets were washed with 1 ml of 75% ethanol, dried at room temperature and dissolved in 50 μl of nuclease-free water. Total RNA concentration was measured with a nanodrop spectrophotometer, and samples were then resolved in 7 M urea 10% acrylamide gel before visualization with SYBR Gold.

Preparation of enriched small RNA fractions from total RNA samples

Cells were grown in 20 ml of medium under the same conditions as those used for the extraction of total RNA. When the induced cultures reached OD_{600} 0.8, cells were centrifuged at 5,000g for 25 min, washed in 5 ml of 0.9% NaCl and centrifuged again at 5,000g for 25 min. The following is adapted from ref. 60. The pellets were suspended in 2 ml of 50 mM sodium acetate (NaOAc) and 10 mM MgOAc pH 5.0, with the addition of 1.9 ml of commercial acidic phenol pH 4.5 (Ambion). The resulting mixture was shaken at 200 rpm for 30 min at 37 °C then centrifuged at 5,000g for 15 min at 4 °C. The upper aqueous phases were collected and subjected to precipitation of total nucleic acids in 0.1 M NaCl, supplemented with an equal volume of 2-propanol, for overnight incubation at room temperature. Following centrifugation at 14,500g for 15 min at 4 °C, pellets were washed with 80% ethanol and air-dried.

Removal of rRNA from samples was achieved through precipitation: pellets were suspended in 0.8 ml of cold 1 M NaCl and spun at 9,500g for 20 min at 4 °C. Supernatants were collected, mixed with 1.7 ml of ethanol, incubated for 30 min at -20 °C and then centrifuged at 14,500g for 5 min at 4 °C. The pellets were then washed with 80% ethanol and air-dried.

Removal of DNA from samples was achieved through precipitation: pellets were resuspended in 0.6 ml of 0.3 M NaOAc pH 5.0. The mixture was then heated for 5 min at 60 °C with regular pipetting, followed by the addition of 0.34 ml of 2-propanol and incubation for 10 min at room temperature; the solution was then centrifuged at 14,500g for 5 min at room temperature and supernatants collected.

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The resulting small RNA fractions were precipitated by the addition of 0.23 ml of ethanol to the supernatant, centrifuging at 14,500g for 15 min at 4 °C, washing the pellet with 80% ethanol and air-drying for 5–10 min. Pellets were dissolved in 0.35 ml of RNase-free water.

For deacylation, the fraction was incubated in 0.1 M Tris pH 9.0 for 45 min at 37 °C. Final precipitation in 0.3 M NaOAc pH 5.0 was achieved by the addition of 2.7 volumes of ethanol, incubation for 30 min at –80 °C and centrifugation at 16,000g for 25 min at 4 °C. Pellets were washed with 80% ethanol, air-dried and dissolved in 80 µl of 1 mM sodium citrate.

Cleavage assay by AriB

Either 50 nM activated AriB or nuclease-deficient AriB (mutant E26A) dimer was incubated for 30 min at 37 °C with 90 ng of enriched small RNAs, extracted as described above, in a 10 µl reaction mix containing 20 mM Tris pH 8.0 and 200 mM NaCl. MgCl₂, when added, was at 2 mM final concentration. Reactions were stopped by the addition of 10 µl of 100% formamide, followed by loading on a 15% acrylamide/7 M urea denaturing medium gel run at 22 W constant. SYBR Gold-stained gels were imaged using the ChemiDoc MP imaging system (Bio-Rad).

Mapping of the AriB cleavage site by primer extension

For precise identification of the AriB cleavage site, DNA probe F693 (5′-FAM/TGGTGGTGGGTCGTGCAGGATTCGAACCTGCGACC-3′, harbouring a fluorescein at the 5′-end) was used in a reaction conducted at room temperature on small RNAs that were incubated with or without activated AriB. In a 12 µl reaction, 2 µl of 1 µM probe was incubated with 750 ng of RNA on a heat block at 95 °C, which was then switched off and left to cool down. Next, 8 µl of PCR with reverse transcription mix (SuperScript III RT, Invitrogen) was added to the reaction mix and loaded onto a thermocycler with the following conditions: 5 min at 25 °C, 60 min at 55 °C and 15 min at 70 °C. Following the addition of 20 µl of 100% formamide, reactions were loaded on a 15% acrylamide/7 M urea denaturing medium gel run at 22 W constant. Fluorescein signal was recorded using the ChemiDoc MP imaging system (Bio-Rad). The ladder was generated by performing an extension reaction using fluorescent probe F693 annealed to a 106 nt DNA template (5′-CGATTGAGGCCGGAATACGACTCACTATAGGGTCGTTAGCTCAGTTGGTAGAGCAGTTGACTTTTAATCAATTGGTCGAGGTTTCAATCCTGCACGACCCACCA-3′) containing the 76 bases corresponding to tRNA^{Lys}. Extension was performed in a Taq DNA polymerase PCR mix incubated under the following conditions: five cycles of 3 min at 95 °C (30 s at 95 °C, 30 s at 66 °C, 2 min at 72 °C), followed by 5 min at 72 °C.

Mapping the AriB cleavage site by specific reverse transcription and Sanger sequencing

AriB-cleaved small RNAs (750 ng) were incubated for 1 h at 25 °C in a 30 µl ligation reaction with 3 µl of 5′ Ligation Reaction Buffer (10X), 2.5 µl of 5′ Ligation Enzyme Mix and 1 µl of 5′ SR Adaptor for Illumina (denatured) from the NEBNext Multiplex Small RNA Library Prep Set (no. E7300S). A tRNA^{Lys}-specific reverse transcription was then performed on 15 µl of the ligation reaction by the addition of 1.5 µl of 10 mM dNTPs, 6 µl of 5× First-Strand Buffer, 1.5 µl of 0.1 M DTT, 1.5 µl of SuperScript III RT from Invitrogen's SuperScript III kit (no. 18080-093) and 50 pmol of the reverse transcription primer (5′-TGTGCTCTCCGATCTGGTGGGTCGTGCAGGATTCGAACCTG-3′) in a total volume of 30 µl, with incubation at 55 °C for 1 h. PCR was performed using the Thermo Scientific DreamTaq Green PCR Master Mix (2×) kit with the following oligos: 5′-AATGATACGGCGACCAACCGAGATCTACACGTTTCAAGTTCTACAGTCCGA-3′ and 5′-CAAGCAGAAGACGGCATACGAGATGTGACTGGAGTTTCAAGCTGTGCTCTTCCGATCT-3′. A band of approximately 120 base pairs was gel purified and sent for Sanger sequencing with the primer 5′-GTGACTGGAGTTTCAAGCTGTGCTCTTCCGATCT-3′.

RNA blot

RNAs were run for 1 h at 180 V in a 7 M urea 10% acrylamide gel and then transferred to a nylon membrane (Invitrogen) using the XCell SureLock Mini-Cell system with the XCell II Blot Module (Invitrogen). The membrane was then cross-linked for 5 min with ultraviolet radiation. Following prehybridization with 30 ml of solution containing 6× SSC (saline sodium citrate buffer), 0.5% SDS and 0.05% casein for 1 h at 45 °C, the membrane was hybridized overnight with 2 nM FAM (final concentration) (6-carboxyfluorescein) or IR (infrared dye IRDye 800 or IRDye 700) labelled oligonucleotide probes (Supplementary Table 3) in 10 ml of prehybridization solution at 45 °C. Finally, the membrane was washed twice with 2× SSC and 0.1% SDS for 10 min at room temperature and twice with 0.2× SSC and 0.1% SDS for 10 min at 60 °C. Images were captured using the ChemiDoc MP Imaging system (Bio-Rad).

The sequences of the *E. coli* tRNA^{Lys} (B803) and T5 tRNA^{Lys} (B806) probes, as well as their homology to *E. coli* and T5 tRNA^{Lys}, are shown in Extended Data Fig. 11a. To confirm the specificity of the T5 tRNA^{Lys} probe (B806), a RNA blot was performed with this probe in the absence of T5 tRNA^{Lys}, showing that our conclusion could not have been affected by non-specific binding of this probe to *E. coli* tRNA^{Lys} or another tRNA (Extended Data Fig. 11b). We can also confirm that the B803 probe used to detect *E. coli* tRNA^{Lys} does not detect T5 tRNA^{Lys}, because the signal of this probe was lost when PARIS was activated in the presence of T5 tRNA^{Lys} (Fig. 5h).

Defence system detection

Defence system detection was performed using DefenseFinder v.1.1.1 (ref. 61) and DefenseFinder model v.1.2.3 on all RefSeq complete genomes from July 2022.

Domain annotation

Defence system domain annotation was done using the HHPred⁶² web-server with PFAM⁶³ database v.33.1 and standard settings. For PARIS, the trigger-binding and coiled-coil domains were annotated using the structure information and ESMfold⁶⁴ structure prediction for AriAB.

AriA and AriB trees

The PARIS systems detected by DefenseFinder v.1.1.1 on RefSeq complete genomes from July 2022 were used to build phylogenetic trees. Only hits to AriA and AriB with a coverage of more than 75% were retained, to avoid pseudogenes. Hits were then clustered using Mmseqs2 (ref. 65) v.13.45111 on AriA and AriAB with an identity threshold of 80% and coverage threshold of 90%. A single representative of each cluster was then used to build the AriA tree, and the cognate AriB was used to build the AriB tree. The AriA tree was built using the AAA15 ATPase domain only and the AriB tree using the DUF4435 domain only. Domains were detected by HMMsearch (from HMMER 3.3.2) on the *E. coli* B185 PARIS system, and a first alignment using Mafft v.7.505 (ref. 66) (default parameters) was used to extract the corresponding sequences in other PARIS variants. The final alignment used to make the tree was carried out using MUSCLE v.5.1.linux64 (ref. 67) using the model super5, and trim using clipkit v.1.3.0 with the option seq-gap. The DUF4435 tree using M5 ribonuclease as outgroup was done using selected representative hits from different clades and aligned using muscle v.5.1.linux64 with the model super5. All trees were built using Iqtree v.2.0.6 (refs. 68,69) using ModelFinder and Ultrafast Bootstrap 1000.

Defence system ATPase AAA15/21 tree

The ATPase AAA15/21 defence system containing detection was performed using hmmsearch from HMMER 3.3.2 and Pfam Hidden Markov Model 33.1 on defence systems previously detected. Random selection of 20 proteins from the detection results with a coverage of more than 75% was used, in addition to experimentally validated homologues, to build the tree. ATPase alignment was carried out using Mafft v.7.505, and

the tree was built using Iqtree v.2.0.6 with ModelFinder and Ultrafast Bootstrap 1000.

Statistics and reproducibility

The pulldown experiments presented in Fig. 3a,b,e,f,i were reproduced in two or three biological replicates. The experiments presented in Fig. 5g,i,j were reproduced three times and in Fig. 5h two times.

Reporting summary

Further information on research design is available in the Nature Portfolio Reporting Summary linked to this article.

Data availability

Sequencing data have been deposited in NCBI database under BioProject ID PRJNA1040033 and GEO accession no. GSE270519. EM maps of PARIS and the atomic model for the asymmetric unit of the complex were deposited to the Electron Microscopy Data Bank (EMDB) and Protein Databank (PDB) databases. Accession codes can be found in Extended Data Table 1. The PDB code for the experimentally determined structure of the PARIS asymmetric unit is 8UX9. EMDB accession code nos. are EMD-42719, EMD-43103, EMD-43104 and EMD-43105. All constructs (WT and mutants) used in this study can be obtained on request to the lead contacts. Source data are provided with this paper.

Code availability

No custom code was used.

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Acknowledgements We thank members of the B.W. laboratory for feedback and discussions, C. Hophan-Nichols and the cyber security team at Montana State University for computational support, H. Callaway for helpful suggestions with data processing and M. Lawrence and C. Gauvin for maintenance and operation of the cryo-EM Core Facility at Montana State University. The cryo-EM Core Facility at Montana State University is supported by NSF no. 1828765 and the M. J. Murdock Charitable Trust. Research in the Wiedenheft laboratory is supported by the National Institutes of Health (no. R35GM134867), the M. J. Murdock Charitable Trust, a young investigator award from Amgen and the Montana State University Agricultural Experimental Station (USDA NIFA). N.B. is supported by an F31 from National Institutes of Health (NIH) (no. GM153146) and received support from Montana INBRE (no. P20GM103474). A.S.-F. is supported by NIH (no. K99GM147842) and the Burroughs Wellcome Fund (no. G-1021106.01). A.B.G. is supported by Montana State University's Undergraduate Scholars Program, and by the NIH NIGMS iDeA programme (no. P20GM103474). J.G., A.T. and R.K. are supported by the European Research Council under the Horizon 2020 Programme (no. 771813). A.I. was supported by RSF grants (nos. 22-14-00004 and 24-74-10089). K.S. was supported by RSF grant 24-14-140018. A.L. was supported by a Ministry of Science and Higher Education grant (no. 075-15-2019-1661). L.S., T.K. and V.H. are supported by the Swedish Research Council (Vetenskapsrådet, no. 2021-01146). We thank A. Glukhov for sharing a collection of T5 deletion variants, T. Prince Maziva for help with in vitro translation assays and A. Demkina for help with BGI sequencing. D.B., F.D., C.R. and B.S. are supported by the European Research Council (no. 101044479) and Agence Nationale de la Recherche (no. ANR-10-LABX-62-IBED). A.B. and F.T. are supported by a European Research Council Starting Grant (no. PECAN 101040529), MSD avenir (UNaDISC project) and the core funding of Institut Pasteur. Molecular graphics and analyses were performed with UCSF ChimeraX, developed by the Resource for Biocomputing, Visualization and Informatics at the University of California, San Francisco, with support from NIH no. R01-GM129325 and the Office of Cyber Infrastructure and Computational Biology, National Institute of Allergy and Infectious Diseases. The funders had no role in the conceptualization, designing, data collection, analysis, decision to publish or preparation of the manuscript.

Author contributions The biochemistry and structural data presented in Fig. 1 were generated by R.A.W., N.B., W.S.H. and A.B.G. The biochemistry and structural data presented in Fig. 2 were generated by R.A.W., N.B., W.S.H. and A.B.G. The plaque presented assays in Fig. 2 were performed by F.D. and S.B. The biochemistry data presented in Fig. 3a,b,e,f,h,i were generated by A.I., S.B., M.S. and A.C. The biochemistry data presented in Fig. 3g were generated by T.Z. The plaque assays presented in Fig. 3d,j were performed by F.D. The structural data presented in Fig. 3c were generated by A.S.-F., R.A.W., N.B., W.S.H. and A.B.G. The efficiency of plating assay presented in Fig. 4a was performed by S.B. and F.D. The light microscopy presented in Fig. 4b was performed by N.M. The metabolic labelling presented in Fig. 4c was performed by L.S., T.K. and V.H. In vitro translation presented in Fig. 4d was performed by S.B., M.S. and A.L. Plaque assays, efficiency of plating and phage challenge assays presented in Fig. 5a–f were performed by A.I., S.B., M.S., F.D. and C.R. The data presented in Fig. 5g–k were performed by F.D. and C.R. Small RNA extractions used in ARII cleavage assays (Fig. 5i) and Sanger sequencing of the cleavage product (Extended Data Fig. 11) were performed by B.S. The evolutionary analysis presented in Fig. 6 was performed by A.B., F.T. and M.B. The Hi-C analysis presented in Extended Data Fig. 9 was performed by R.K., J.G. and A.T. Project conceptualization, administration and the original draft were undertaken by B.W., A.I. and D.B., with review and editing by all authors.

Competing interests The authors declare no competing interests.

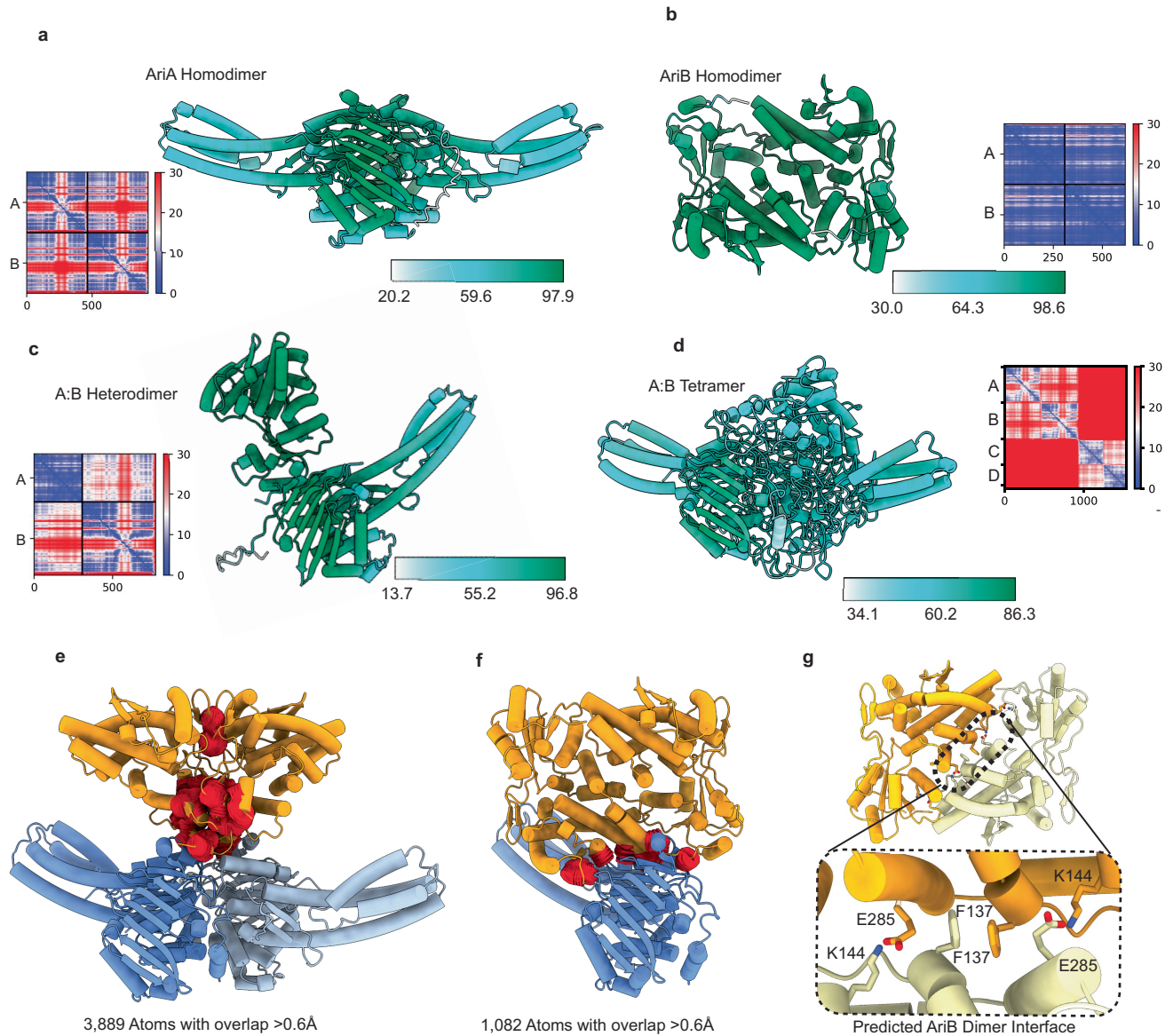
Additional information

Supplementary information The online version contains supplementary material available at <https://doi.org/10.1038/s41586-024-07874-3>.

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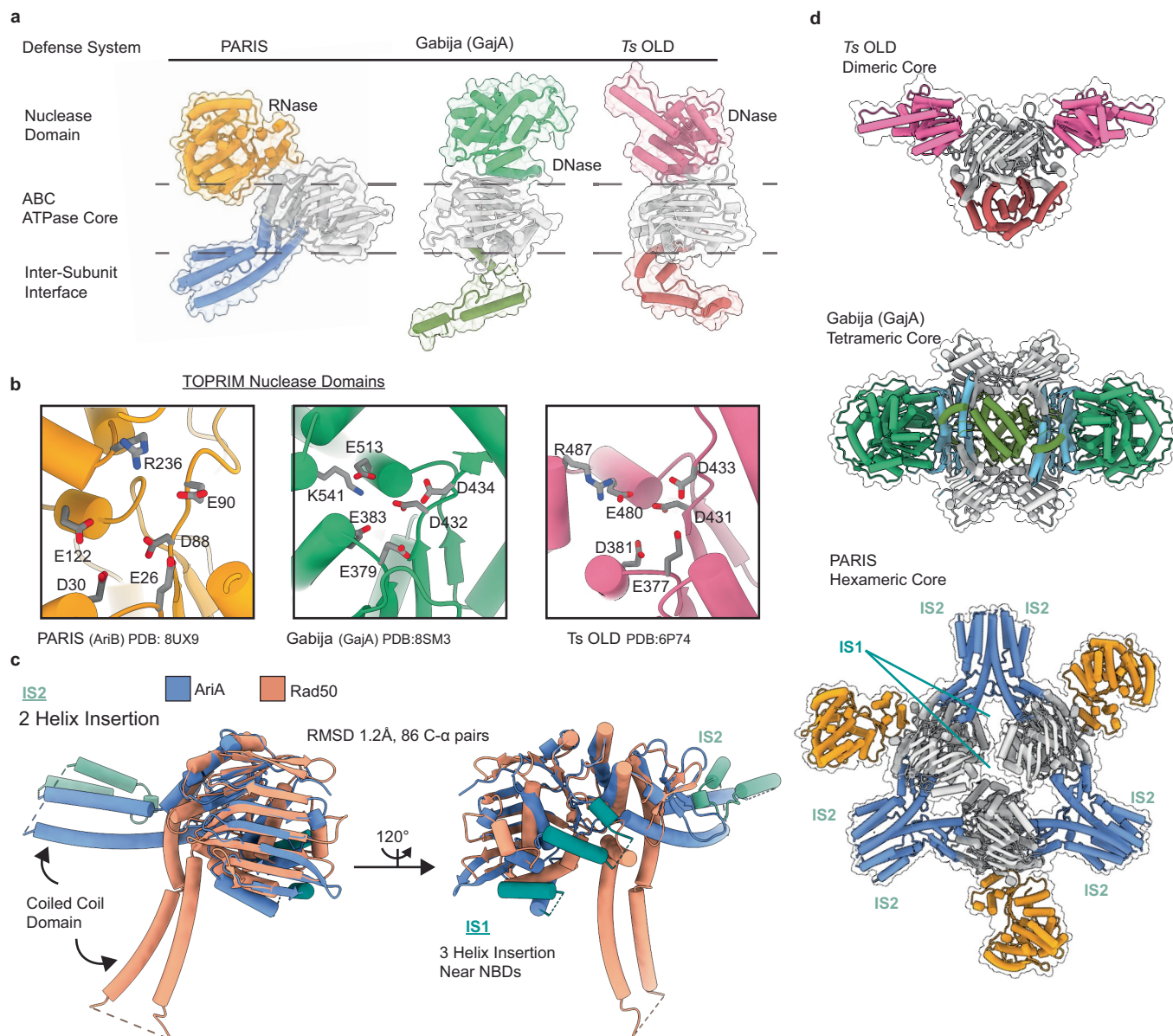
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Extended Data Fig. 1 | AlphaFold2 structural predictions of AriA and AriB. AlphaFold2-predicted structures of AriA and AriB oligomers colored by the predicted local distance difference test (pLDDT) score (0–100) with associated scale bars. Values greater than 90 indicate high confidence and values below 50 indicate low confidence. Predicted Aligned Error plots (PAE) are also shown for each structural prediction. **a**, AlphaFold2-predicted structure of the AriA homodimer. Confidence is high (PLDDT >90, green) for the ATPase domain, and low for the insertion sequences (PLDDT <70, teal). **b**, AlphaFold2-predicted structure of the AriB homodimer results in a high confidence model with more >98% of the residues above a PLDDT of 90. **c**, AlphaFold2-predicted structure of

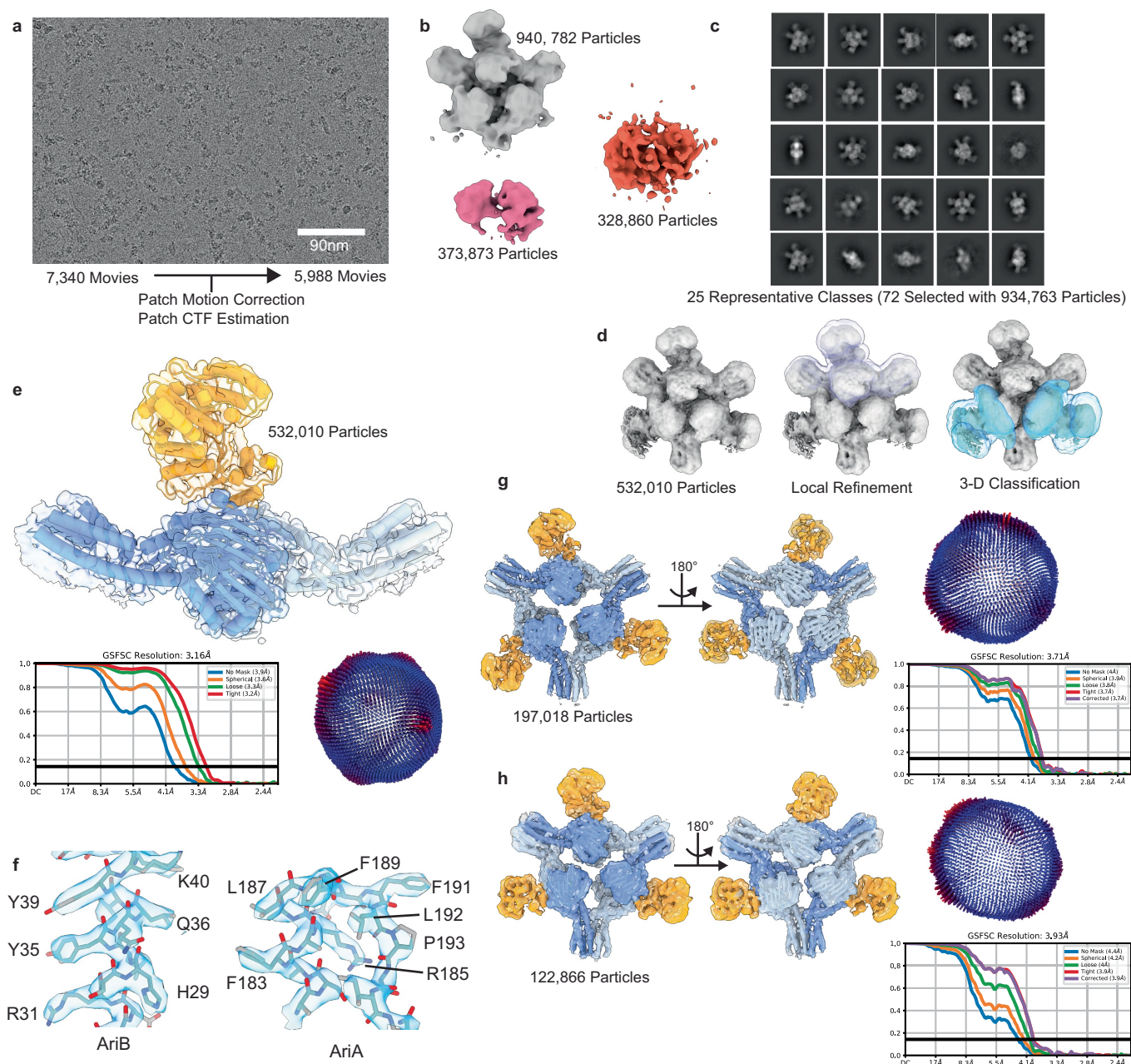
the AriA-B heterodimer and associated PAE plot. **d**, AlphaFold2 returned unreliable models when attempting to predict structures for higher ordered assemblies of AriA and AriB as demonstrated by the PLDDT plot below 50 for most of the molecule. **e**, Clashing (red) between AriBs, prevents the assembly of two AriB subunits on a single AriA homodimer. Clashing is defined by atoms with overlap greater than 0.6 Å. **f**, A predicted homodimer of AriB cannot associate with a homodimer of AriA without clashing (red). Atoms with an overlap greater than 0.6 Å are highlighted with red disks signifying the clash. **g**, Predicted structure of the AriB dimer shown in panel b, highlighting residues involved in dimerization.



Extended Data Fig. 2 | Structural comparison of PARIS homologs.

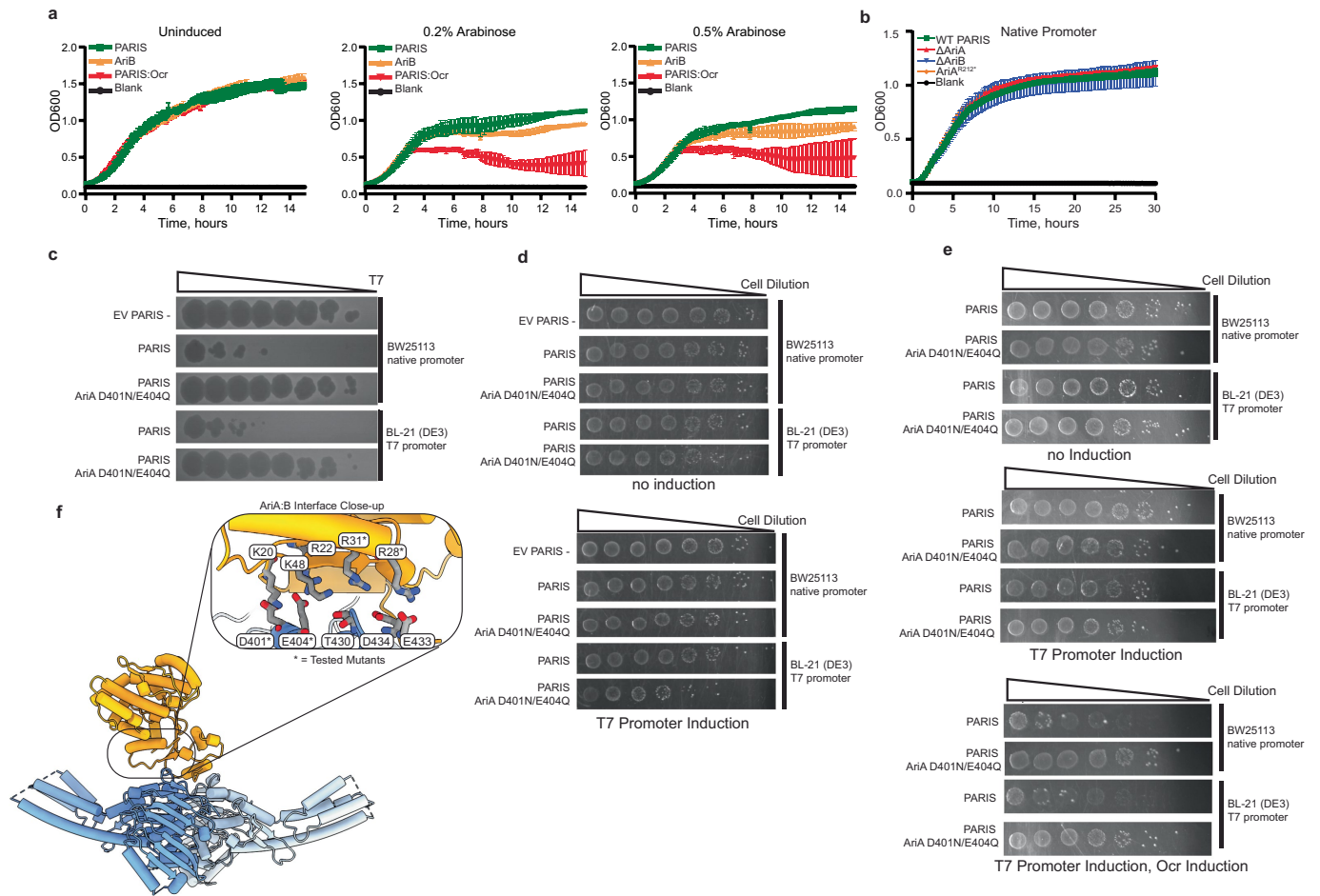
a, Domain organization of phage defense systems where an ABC ATPase domain is associated with a TOPRIM nuclease domain. While the ATPase domain is highly conserved, the relative orientation of the TOPRIM nuclease domains and dimerization domains vary between PARIS (PDB ID: 8UX9), Gabija (PDB ID: 8SM3), and the *Thermus scotoductus* Overcoming Lysogenization Defect (*Ts* OLD, PDB ID: 6P74). **b**, AriB is the effector of PARIS defense. The AriB TOPRIM is homologous to the TOPRIM domains of *Ts* OLD (RMSD 1.1 Å, 20 C-α pairs) and GajA (RMSD 1.02 Å, 26 C-α pairs). The TOPRIM domain

of AriB is within the DUF4435, which plays a role in AriB dimerization. **c**, Structural comparison of AriA to Rad50, a universally conserved ATPase. AriA primarily differs from Rad50 at Insertion Sequence (IS) 1 (residues 122-180) and IS2 (residues 242-289). IS1 introduces three alpha helices near the nucleotide binding domains of AriA that are predicted to be involved in trigger recognition. IS2 expands the coiled coil domain with two alpha helices and enable the homo-hexameric assembly of AriA subunits in the PARIS complex. **d**, Structural comparison of PARIS and related OLD systems highlights the unique assembly state of PARIS.



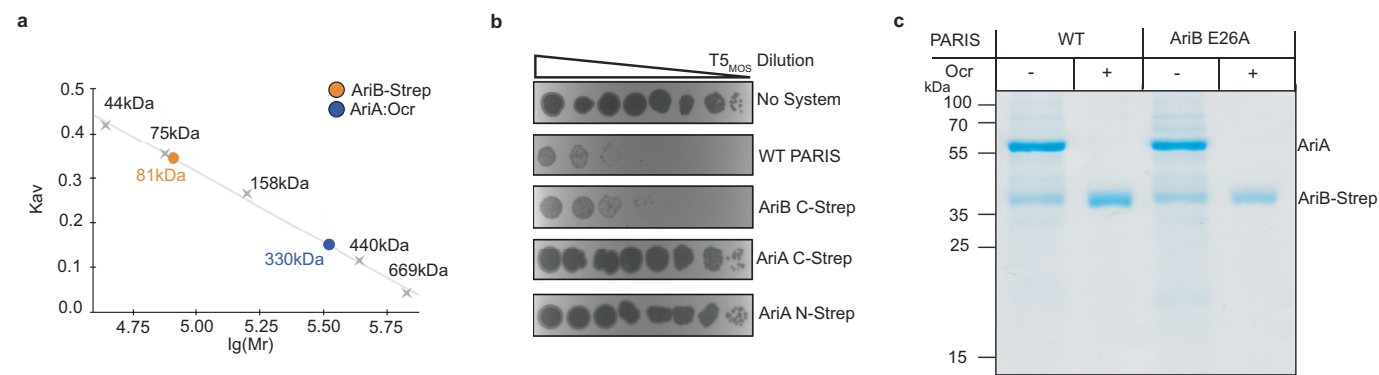
Extended Data Fig. 3 | Workflow for structural determination of the PARIS complex. Image processing was performed using cryoSPARC v4.41. **a**, 7,340 movies were recorded. Micrographs with CTF-fits worse than 8 Å were removed prior to downstream processing ($n = 5,988$). Using a *de novo* template generated from a 200-micrograph subset of this data, 4,078,384 particles were identified and extracted. From 17 selected classes, 1,643,515 particles remained, and a 3-class *ab initio* reconstruction yielded volumes shown in panel **b**. Particles belonging to junk classes (pink and red) were discarded. **c**, 25 representative 2-D Classes from 940,782 particles in panel **b**. A total of 934,763 particles from 72 classes remained after final 2-D Classification ($n = 100$ classes). **d**, After successive rounds of *ab initio* and heterogenous refinements, a 532,010 particle

volume was obtained, corresponding to the intact PARIS complex with compositional heterogeneity at two AriB attachment sites. Masks were generated for local refinement and 3-D classification to produce the reconstructions shown in **e**, **g**, and **h**, respectively. **e**, Local Refinement of one asymmetric unit of the PARIS immune complex with FSC and 3-D orientation plot shown below. **f**, Close up of density map and model from **e** demonstrating map quality. **g**, C3 reconstruction of the PARIS complex at 3.71 Å-resolution with the experimental structure determined in panel **e** fit into the density. **h**, Cryo-EM reconstruction of PARIS with AriB in the 'trans' arrangement at approximately 3.93 Å-resolution with experimental structure determined in panel **e**, fit into the density.



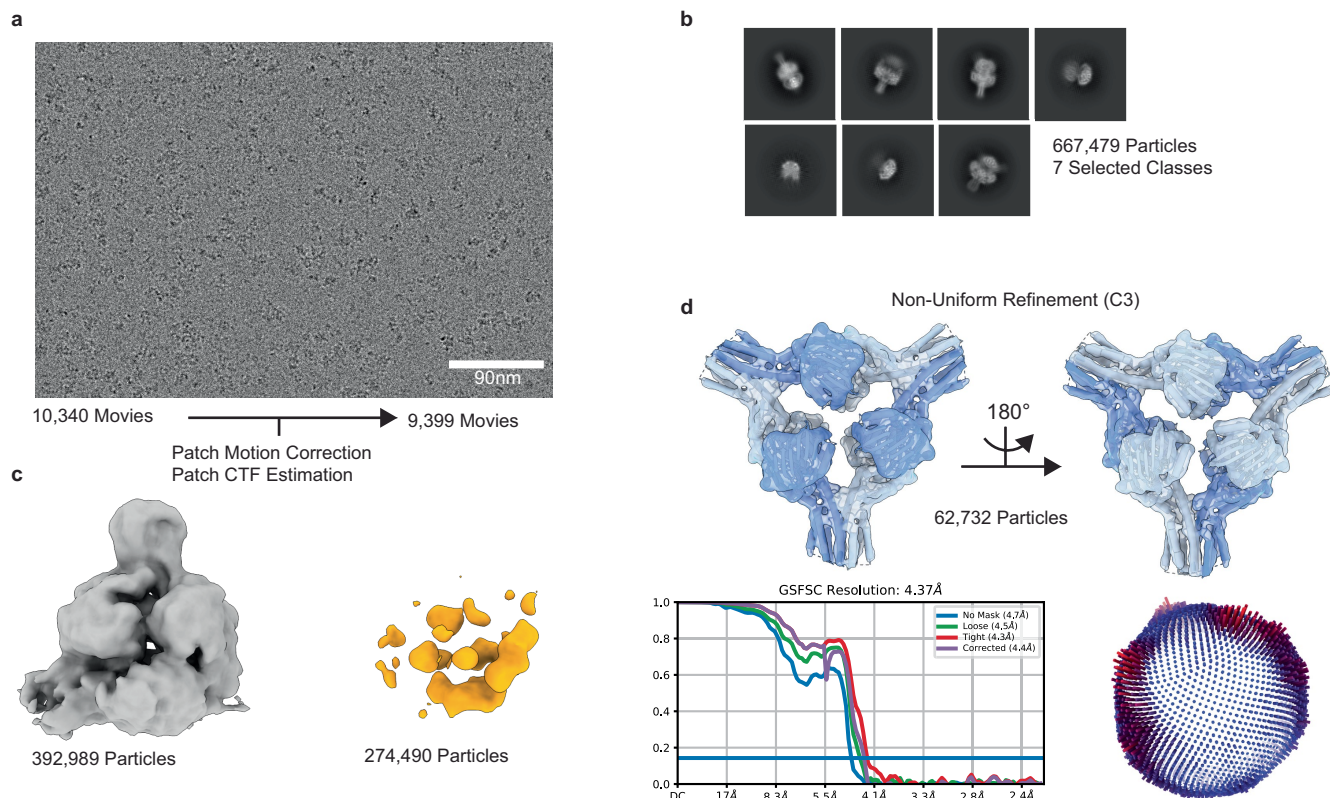
Extended Data Fig. 4 | AriB toxicity is dependent on association with AriA.
a, Liquid culture toxicity assays of AriB alone or AriA:B together (PARIS). Genes are overexpressed using a T7 inducible promoter. AriB alone results in a mild growth defect, while Ocr-triggered release of AriB (PARIS Ocr) is significantly more toxic. **b**, Growth of *E. coli* MG1655 carrying plasmid pFR85 (pSC101 with *ariAB* under the control of the native promoter), or variants with *ariA* deleted, *ariB* deleted or a stop codon in *ariA* yielding an interrupted protein that lacks the interaction interface with AriB. **c**, Structure-guided mutations that disrupt

the AriA:AriB interface result in a loss of defense phenotype. **d**, Drop-test toxicity assay with PARIS (WT) or PARIS with AriA D401N/E404Q mutation. A modest growth defect can be observed only for the PARIS AriA D401N/E404Q overexpressed from T7 promoter. **e**, The addition of pBAD Ocr to the PARIS AriA mutants does not increase the toxicity, indicating that AriB is not functional when AriA interactions are disrupted. **f**, Close-up of the AriA:AriB interaction interface. Mutated residues are labeled with asterisks.



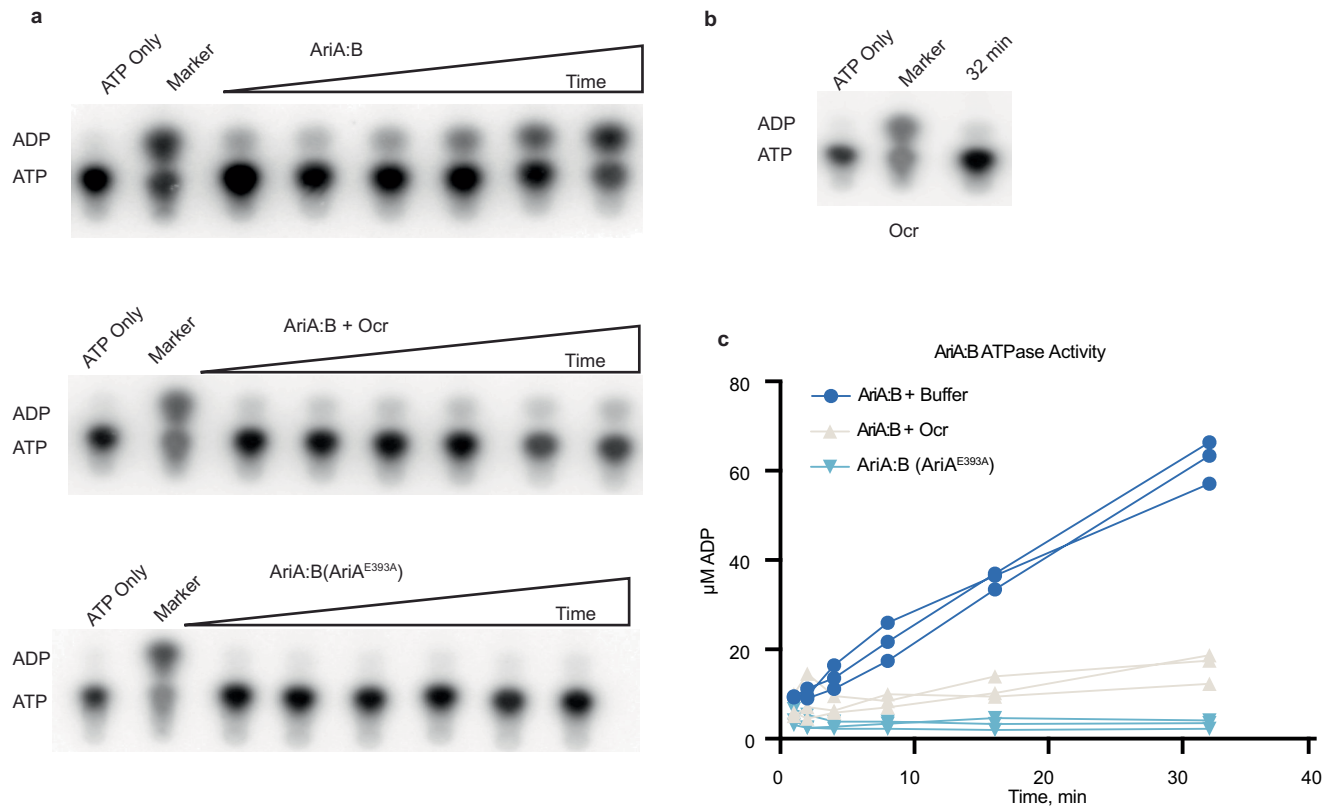
Extended Data Fig. 5 | Purification of activated AriB. a, Size exclusion chromatography calibration curve with estimated masses of AriA:Ocr and activated AriB complexes. **b**, Introduction of the Strep-tag at the C-

or N- terminus of AriA, but not on the C-terminus of AriB, interferes with PARIS anti-phage defense. **c**, Purification of activated AriB after mixing cell lysates containing strep-tagged AriB and non-tagged Ocr.



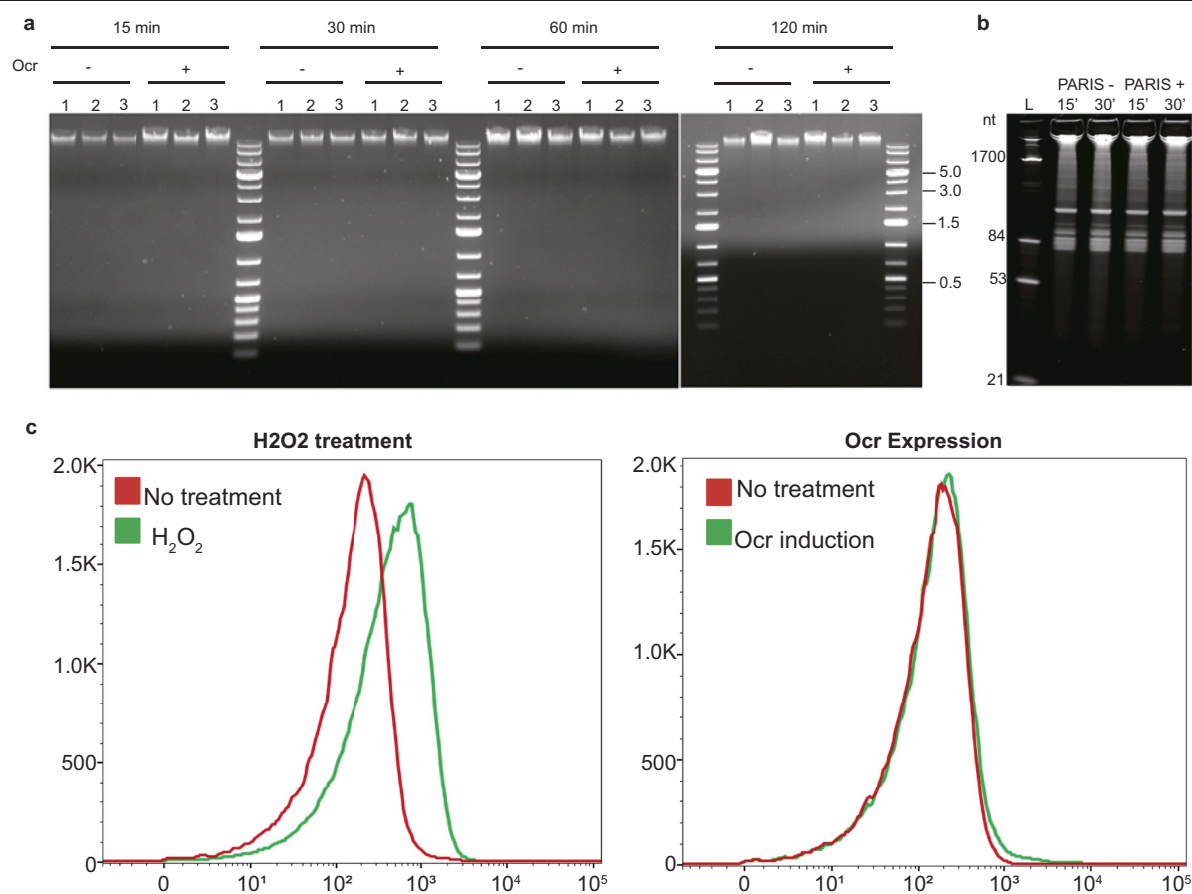
Extended Data Fig. 6 | Structure determination of AriA purified using a Ocr-Strep pulldown. Image processing and analysis was performed in cryoSPARC v4.41. **a**, Representative micrograph from a total of 10,340 movies. Micrographs with CTF-fits worse than 8 Å-resolution were discarded. Particle picking on 9,399 micrographs identified 4,415,782 particles which were extracted and subjected to 2-D Classification. **b**, Selected 2-D classes corresponding to the AriA hexamer contained 667,479 particles. **c**, 2-Class *ab initio* reconstruction of particles from panel **b**. The gray class corresponds

to the AriA homohexamer, while particles associated with the orange class were discarded. Density for the ATPase domains is present, but one blade of the propeller is poorly resolved due to the increased flexibility of the AriA scaffold relative to the fully assembled PARIS complex. **d**, After iterative rounds of sorting the 392,989 particles from panel **c**, a stack containing 62,732 particles was isolated and C3 refinement produced a 4.4 Å-resolution reconstruction of the AriA scaffold.



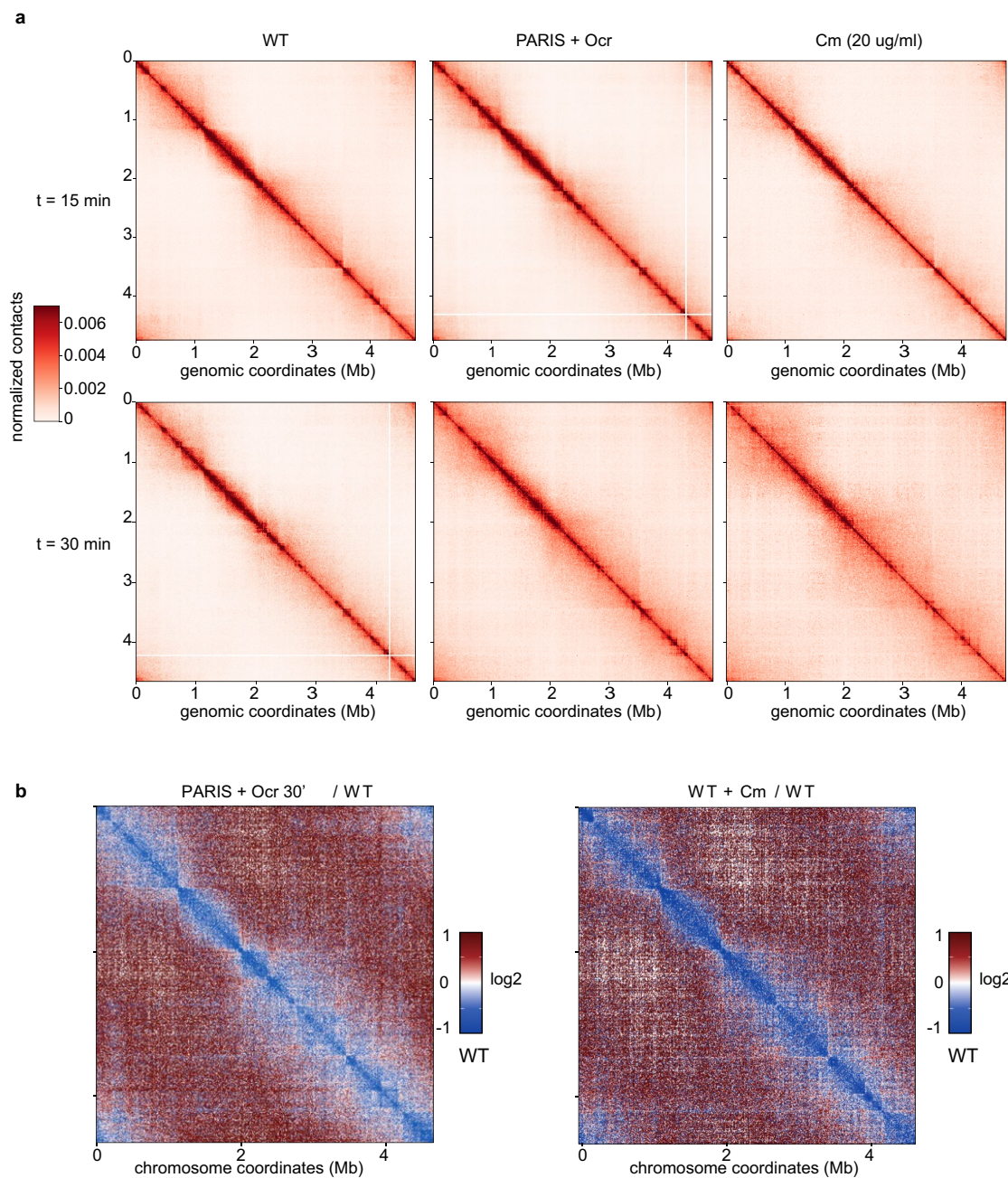
Extended Data Fig. 7 | AriA ATPase activity. **a**, PARIS mediated hydrolysis of $\alpha^{32}\text{P}$ ATP with PARIS alone or with 10-fold excess trigger (T7 Ocr). Reactions were run from 1 to 32 min and products were resolved via thin layer chromatography. Reactions with $\alpha^{32}\text{P}$ ATP incubated with T4 polynucleotide kinase were used as a positive control. Images are representative of reactions that were performed in

triplicate. **b**, Ocr alone was incubated with $\alpha^{32}\text{P}$ ATP for 32 min indicating that the trigger alone does not hydrolyze ATP. **c**, Concentration of ADP formed at each time point was quantified and plotted. Data points of three technical replicates are shown.



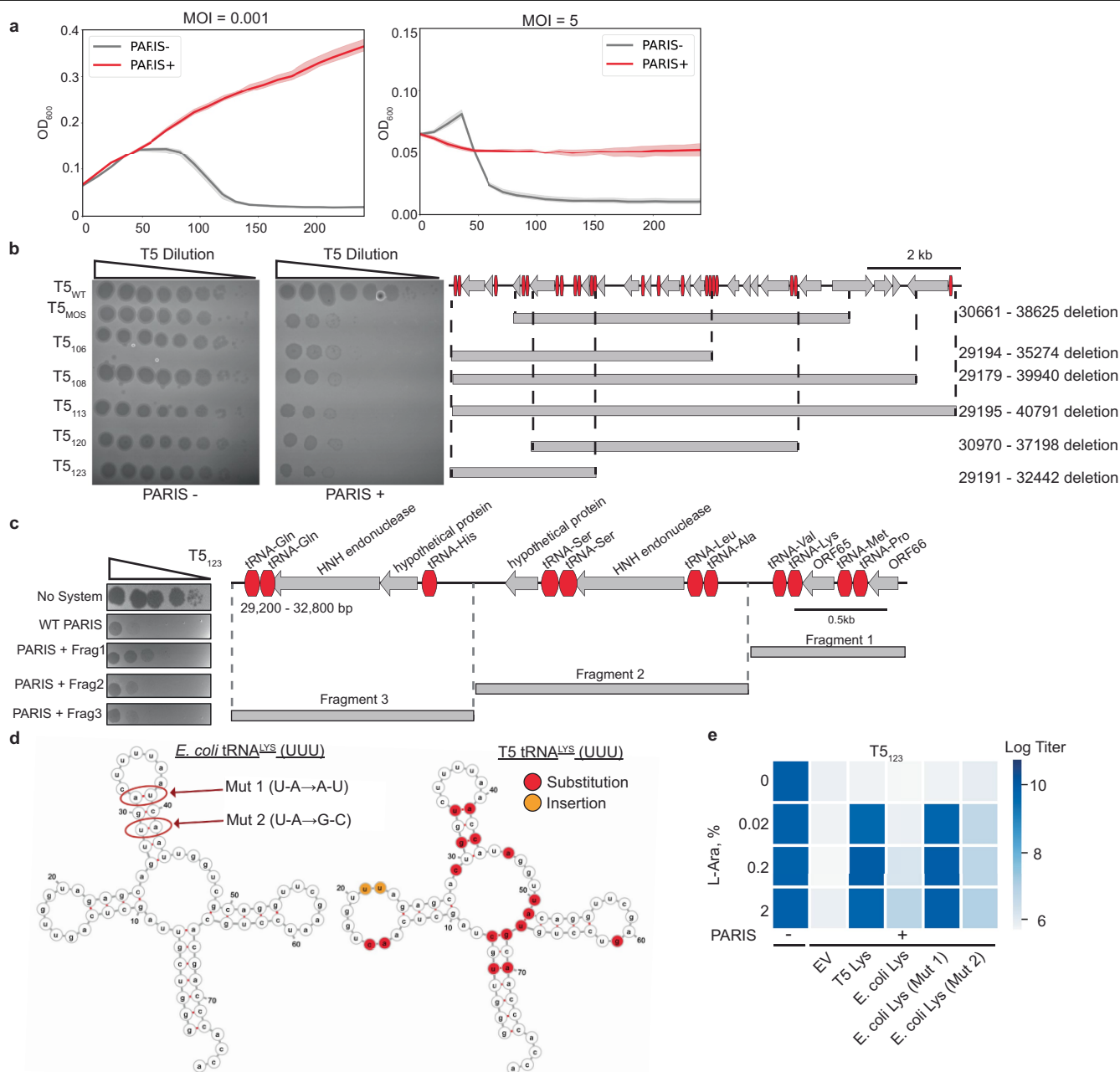
Extended Data Fig. 8 | PARIS activation does not result in total RNA or DNA degradation. **a**, Total DNA was extracted from *E. coli* MG1655 carrying plasmids pFR85 (PARIS) and pFD250 (P_{PhIF}-Ocr) at different time points after Ocr induction with or without DAPG (n = 3), or from a non-induced control. Samples were run on a 1% agarose TAE gel. **b**, Total RNA was extracted from *E. coli* MG1655 carrying plasmid pFR85 (PARIS) or pFR66 (sfGFP), and pFD250 (P_{PhIF}-Ocr) at different time points after induction of Ocr expression with DAPG. Samples

were run on a TBE Urea (7M) acrylamide (10%) gel and stained with SYBR-Gold. A representation of 3 replicates is shown. **c**, TUNEL assay, demonstrating the lack of accumulation of dsDNA breaks in PARIS⁺ cells 2 h after Ocr induction, as measured through terminal deoxynucleotidyl transferase labeling of free 3'-OH groups in DNA. As a positive control of DNA damage, cells were treated with 0.1% H₂O₂.



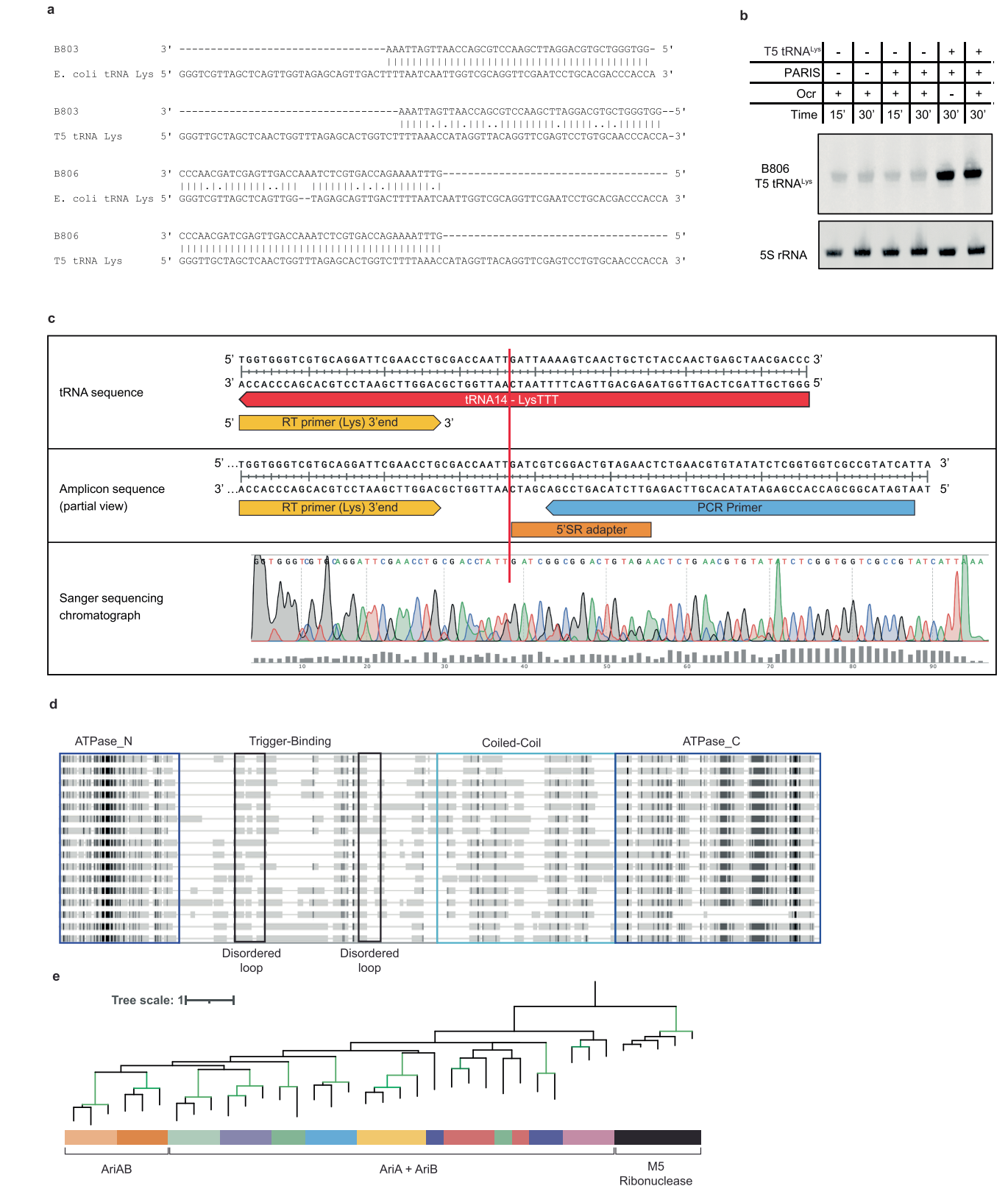
Extended Data Fig. 9 | PARIS activation results in DNA compactization akin to inhibition of translation with chloramphenicol. a, Hi-C contact maps (bin: 4kb). From left to right: WT (pFR66 + pFD245 control vectors), PARIS (pFR85) + P_{PHIF}-ocr (pFD250), and chloramphenicol treated cells. Top and

bottom rows correspond to 15 and 30 min after induction with DAPG or treatment with chloramphenicol. **b**, Ratio between contact maps of WT at t = 30 min and contact maps 30 min after either PARIS activation (left) or chloramphenicol treatment (right).



Extended Data Fig. 10 | Identification of the T5 genomic region required for infection of PARIS⁺ cells. **a**, Growth of PARIS⁻ or PARIS⁺ cells infected with PARIS-sensitive phage T5_{MOS} at low or high MOI. The phage was added at $t = 0$. **b**, Phage T5 variants with non-essential deletions in the region encoding tRNAs are sensitive to PARIS defense. Boundaries of the deletions are shown on the right, tRNA genes are highlighted in red. **c**, Overexpression of Fragment 1

(31885–32870) derived from the deletion in the phage T5₁₂₃ partially rescues T5₁₂₃ infection of the PARIS⁺ culture. **d**, Comparison of *E. coli* and T5 tRNA^{Lys} highlights substitution mutations near the anticodon stem loop that we hypothesize prevents targeting by AriB. **e**, Swapping U-A for A-U (Mut1) on the *E. coli* tRNA^{Lys} confers resistance to PARIS.



Extended Data Fig. 11 | tRNA^{Lys} cleavage product and phylogenetics of PARIS. **a**, Alignment of the B803 and B806 Northern Blot probes to the *E. coli* and T5 tRNA^{Lys}. **b**, Northern Blot with probe B806 in the presence or absence of the T5 tRNA^{Lys} expressed from a pBAD. **c**, Mapping of the tRNA^{Lys} cleavage site by reverse-transcription, adapter ligation, PCR and Sanger sequencing. The cleavage site, identified as the junction between the tRNA sequence and the

ligated adapter, is marked with a vertical red line. **d**, Alignment of homologs of AriA representative of the different PARIS clades. Different domains of AriA are indicated. Grey scale represents % of identity between homologs. **e**, Phylogenetic tree of DUF4435 domain of AriB using M5 Ribonuclease (TOPRIM) as an outgroup.

Extended Data Table 1 | Cryo-EM data collection, processing, and model validation

	PARIS Asymmetric Unit (PDB ID: 8UX9, EMD-42719)	PARIS C3 Arrangement (EMD-43103)	PARIS 'trans' Arrangement (EMD-43104)	AriA-Ocr (EMD-43105)
Data Collection and Processing				
Microscope	Talos Arctica			
Voltage	200kV			
Camera	Gatan K3			
Magnification	36,000 X			
Pixel size (Å)	0.552	0.552	0.552	0.552
Total Dose ($e/\text{Å}^2$)	56.42	56.42	56.42	56.31
Exposure time (s)	4.993	4.993	4.993	4.107
Frames/Exposure	50	50	50	50
Defocus Range (μm)	-0.6, -2.5	-0.6, -2.5	-0.6, -2.5	-0.6, -2.5
Number of Micrographs	7,340	7,340	7,340	10,034
Number of Initial Particles	4,078,384	4,078,384	4,078,384	667,479
Number of particles for 3-D Analysis	532,010	532,010	532,010	100,862
Symmetry	C1	C3	C1	C3
Number of Final Particles	532,010	197,018	122,866	62,732
Resolution (0.143 FSC, Å)	3.2	3.7	3.9	3.7
Model Building and Refinement Statistics				
Refinement Software	Phenix. real space refine			
Clash score	5.75			
MolProbity score	1.71			
Poor rotamers	24 (4%)			
Favored rotamers	529 (96%)			
Ramachandran outliers	1 (.09%)			
Ramachandran Favored	1019 (94%)			
Bad bonds	2			
Bad Angles	4			

This table lists the refinement statistics for cryo-EM structures of PARIS.

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For all statistical analyses, confirm that the following items are present in the figure legend, table legend, main text, or Methods section.

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☐	☒ A statement on whether measurements were taken from distinct samples or whether the same sample was measured repeatedly
☐	☒ The statistical test(s) used AND whether they are one- or two-sided <i>Only common tests should be described solely by name; describe more complex techniques in the Methods section.</i>
☒	☐ A description of all covariates tested
☐	☒ A description of any assumptions or corrections, such as tests of normality and adjustment for multiple comparisons
☐	☒ A full description of the statistical parameters including central tendency (e.g. means) or other basic estimates (e.g. regression coefficient) AND variation (e.g. standard deviation) or associated estimates of uncertainty (e.g. confidence intervals)
☐	☒ For null hypothesis testing, the test statistic (e.g. <i>F</i> , <i>t</i> , <i>r</i>) with confidence intervals, effect sizes, degrees of freedom and <i>P</i> value noted <i>Give P values as exact values whenever suitable.</i>
☒	☐ For Bayesian analysis, information on the choice of priors and Markov chain Monte Carlo settings
☒	☐ For hierarchical and complex designs, identification of the appropriate level for tests and full reporting of outcomes
☒	☐ Estimates of effect sizes (e.g. Cohen's <i>d</i> , Pearson's <i>r</i>), indicating how they were calculated

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Software and code

Policy information about [availability of computer code](#)

Data collection	SerialEM v3.8.6
Data analysis	cryoSPARC v4.41, DeepEMhancer v0.14, ISOLDE v1.7.1, PHENIX v1.20.1, COOT v0.9.8.1, ChimeraX v1.7, bowtie2 v2.4.4, hicstuff v3.0.3, GraphPad Prism 10.1.2, FlowJo v10, Unicycler v0.5.0, DefenseFinder v1.1.1, DefenseFindermodels v1.2.3, Mmseqs2 v12.54111, HMMER 3.3.2, MAFFT v7.505, MUSCLE v5.1, clipkit v1.3.0, Iqtree v2.0.6

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All information is available upon request to the lead contacts. Sequencing data have been deposited in NCBI database under BioProject id PRJNA1040033. EM maps of PARIS in various assembly states and the atomic model for the asymmetric unit of the complex were deposited to the Electron Microscopy Data Bank

(EMDB) and Protein Databank (PDB) databases. Accession codes can be found in Supplementary Table 1 of the manuscript. The PDB code for the experimentally determined structure of the PARIS asymmetric unit is 8UX9. EMD Accession codes for the maps produced in this study are available under EMD-42719, EMD-43103, EMD-43104, and EMD-43105. Raw micrographs are uploaded to Electron Microscopy Public Image Archive (EMPIAR) under the accessions EMPIAR-11832 and EMPIAR-11833. HMMs used for defense system annotation are available in the PFAM database v33.1. All constructs (wild-type and mutants) used in this study can be obtained on request to the lead contacts.

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Population characteristics N/A

Recruitment N/A

Ethics oversight N/A

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Life sciences study design

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Sample size	No sample size calculations were performed as this is not relevant to the structural, biochemical, and phage assay data reported in this study. For cryo-EM analysis, final image numbers for each dataset were collected to obtain sufficient EM maps that allow for unambiguous atomic model building which indicates sufficient sample size.
Data exclusions	During Cryo-EM data collection, micrographs that failed to meet certain quality criteria (CTF fit) were discarded along with "junk" particles that failed to align to high resolution. This is standard practice in Cryo-EM datasets. No other data were excluded.
Replication	Experiments were performed with replicates as described. Toxicity and phage challenge assays were performed with 3 biological replicates. For toxicity assays, the mean value of replicates was plotted with error bars reflecting standard deviation. For phage challenge assays, the result of one biologically independent replicate are shown and the replicates not shown reflect results within expected variation. Numbers of replicates for relevant experiments are described in the methods.
Randomization	Division of particles into random halves is automatically carried out during 3D reconstruction in cryoSPARC. Randomization is not relevant to other biochemical analyses.
Blinding	Blinding is not relevant to this study as the structural and biochemical results are not subjective.

Reporting for specific materials, systems and methods

We require information from authors about some types of materials, experimental systems and methods used in many studies. Here, indicate whether each material, system or method listed is relevant to your study. If you are not sure if a list item applies to your research, read the appropriate section before selecting a response.

Materials & experimental systems

n/a	Involvement in the study
☒	☐ Antibodies
☒	☐ Eukaryotic cell lines
☒	☐ Palaeontology and archaeology
☒	☐ Animals and other organisms
☒	☐ Clinical data
☒	☐ Dual use research of concern
☒	☐ Plants

Methods

n/a	Involvement in the study
☒	☐ ChIP-seq
☒	☐ Flow cytometry
☒	☐ MRI-based neuroimaging

Plants

Seed stocks

N/A

Novel plant genotypes

N/A

Authentication

N/A